Sinais e Sistemas para Engenharia

MÓDULO 7 - ANÁLISE DE SISTEMAS POR LAPLACE

VERSÃO OCTAVE

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Resposta de Sistemas à sinais contínuos

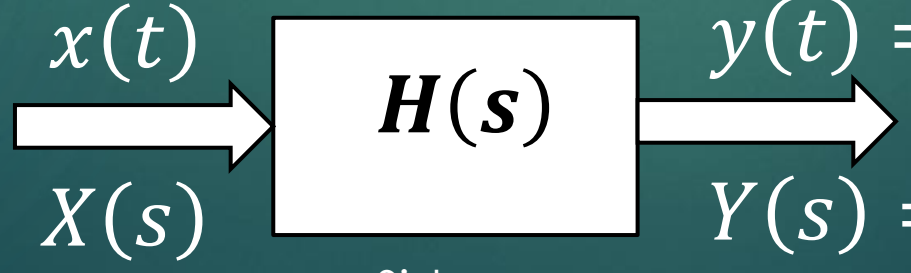
Os sistemas físicos são normalmente representados por equações diferenciais. Estes sistemas são excitados por sinais/funções contínuos. Há 3 opções para se calcular a resposta $y(t)$ do sistema a uma entrada $x(t)$:

1) Resolver a equação diferencial que define o sistema, opção que já estudamos

2) Calcular a resposta do sistema através da convolução do sinal de entrada com a resposta ao impulso do sistema, opção de já estudamos

$$y(t) = x(t) * h(t)$$

3) A opção que iremos aprofundar agora será calcular primeiro $Y(s)$ (transformada de Laplace da saída) através da multiplicação de $X(s)$ (transformada de Laplace da entrada) por $H(s)$ (função de transferência do sistema). A saída $y(t)$ será a transformada inversa de Laplace de $Y(s)$.

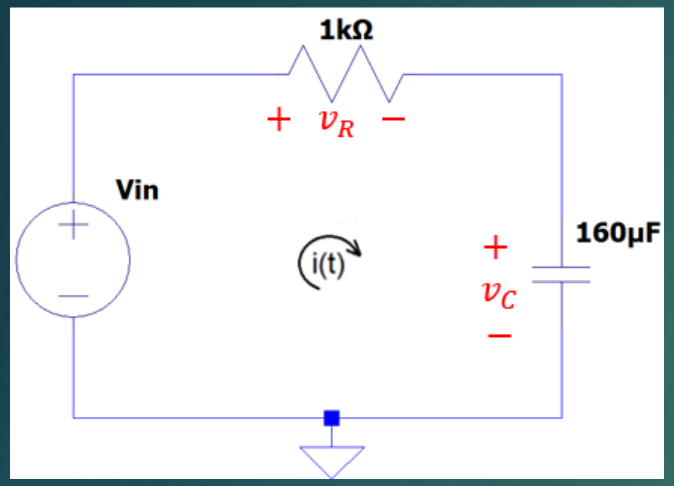


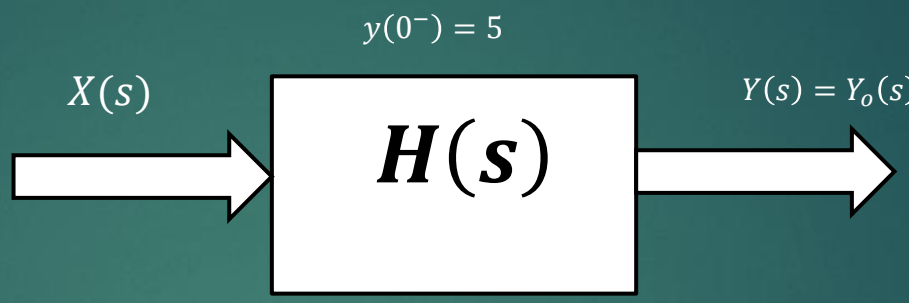
$$y(t) = x(t) * h(t)$$

$$Y(s) = X(s)H(s)$$

Sistemas Elétrico de 1ª Ordem

Como exemplo, seja o sistema eletrônico R mostrado abaixo no qual se quer encontrar a tensão no capacitor $v_c(t)$ (saída do do sistema) , resultante da tensão na fonte $v_{in}(t)$ (entrada do sistema)





$$Y(s) = Y_o(s) + Y_n(s) = X(s)H(s) + X_n\left(\frac{1}{H(s).den}\right)$$

$x(t)$ – fonte de tensão

$y(t)$ – tensão no capacitor

Este sistema elétrico é definido pela seguinte equação diferencial:

$$y'(t) + \frac{1}{\mathrm{RL}}y(t) = \frac{1}{RC}x(t)$$

$$y'(t) + 6.25y(t) = 6.25x(t)$$

$R = 1k\Omega$

$C = 160\ \mu F$

A entrada é representada pelo sinal $\mathrm{i_s(t)}$ e a saída pelo sinal $\mathrm{i_L(t)}$

Condições Iniciais

$y(0^-) = 5V$

$y'(0^-) = 0$

Função de Transferência do Sistema

A função de transferência no domínio de Laplace é definida como sendo a razão entre a transformada de Laplace do sinal de saída pela transformada de Laplace do sinal de entrada

Condições iniciais nulas

$y'(t) + 6.25y(t) = 6.25x(t)$

Aplicacando Laplace em ambos os lados

$sY(s) + 6.25Y(s) = 6.25X(s)$

$(s + 6.25)Y(s) = 6X(s)$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{6.25}{(s + 6.25)}$$

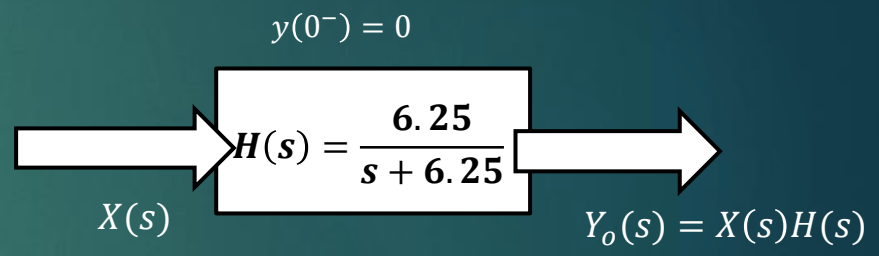
$Y_o(s) = X(s)H(s)$

$$\boxed{H(s) = \frac{Y(s)}{X(s)} = \frac{\mathbf{6.25}}{\boldsymbol{s + 6.25}}}$$

Entrada	$x(t) \xrightarrow{\mathcal{L}} X(s)$
Saída	$y(t) \xrightarrow{\mathcal{L}} Y(s)$
	$y'(t) \xrightarrow{\mathcal{L}} sY(s)$

$$\boldsymbol{H_n(s)} = \frac{\mathbf{1}}{\boldsymbol{H(s).den}} = \frac{\mathbf{1}}{\boldsymbol{s + a}}$$

$X_n(s) = y(0)$



Com Condições Iniciais

$y'(t) + 6.25y(t) = 6.25x(t)$

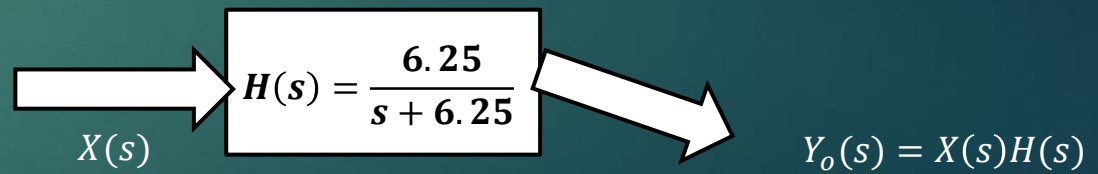
$(sY(s) - y(0)) + 6.25Y(s) = 6.25X(s)$

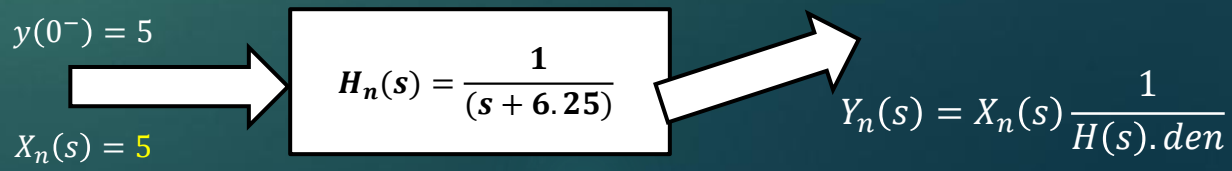
$(s + 6.25)Y(s) - y(0) = 6.25X(s)$

$$Y(s) = X(s)\frac{6.25}{(s + 6.25)} + (y(0))\frac{1}{(s + 6.25)}$$

$$Y(s) = X(s)H(s) + X_n(s)\left(\frac{1}{H(s).den}\right)$$

$$Y(s) = Y_o(s) + X_n(s)\left(\frac{1}{H(s).den}\right)$$





Calcular a Resposta ao impulso $\mathrm{h}(t)$ do Sistema: $x(t) = \delta(t)$

Condições Iniciais Nulas

$X(s) = 1$ → $\boldsymbol{H(s) = \dfrac{6.25}{s + 6.25}}$ → $Y(s) = X(s)H(s) = H(s)$

Resposta ao impulso

Como $x(t) = \delta(t) \xrightarrow{\mathcal{L}} X(s) = 1$

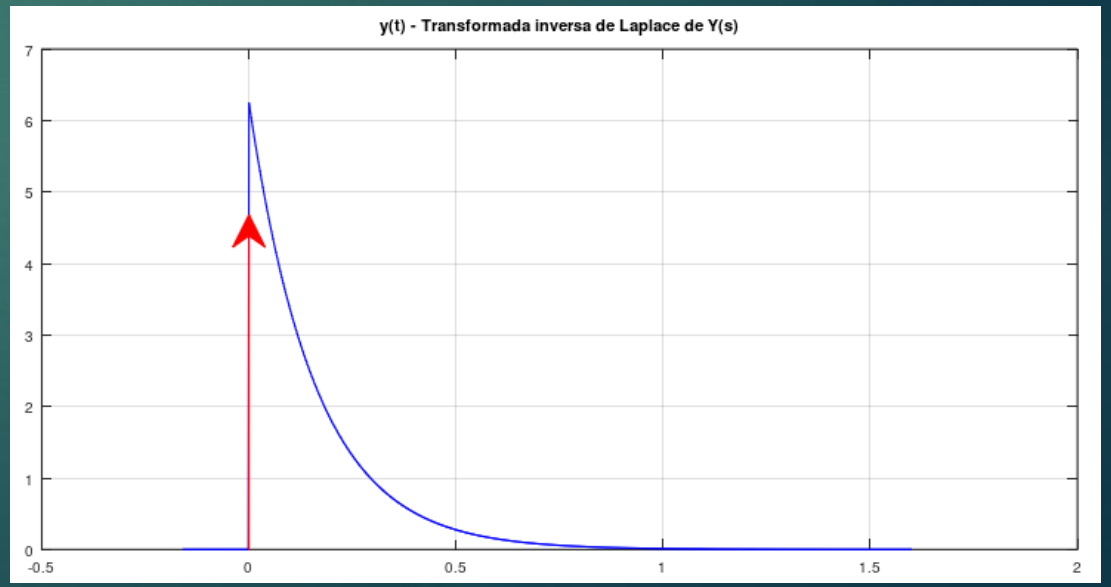
$$Y(s) = X(s)H(s) = (1)H(s) = \frac{6.25}{(s + 6.25)}$$

Então a resposta $h(t)$ do sistema ao impulso será a transformada inversa de Laplace da própria função de transferência

$$H(s) = \frac{6.25}{(s + 6.25)} \xrightarrow{\mathcal{L}^{-1}} h(t) = 6.25\mathrm{e}^{-6.25\mathrm{t}}\mathrm{u(t)}$$

```
>> Hs=6.25/(s+6.25);
>> Xs=1;
>> ft=ilaplace_fga(Hs*Xs,true);
ft=@(t)(+(6.25000*exp(-6.25000*t)) ).*f_u(t);
```

```
>> RespostaSistemaHs(Hs,1,0,1,true)
--Saida--
ft=@(t)(+(6.25000*exp(-6.25000*t)) ).*f_u(t);

--Entrada--
fx=@(t)(y=(1.00000)*f_d(t));
```



Calcular a Resposta ao Degrau $y(t)$ do Sistema: $x(t) = u(t)$

O índice '0' em $Y_o(s)$ e $y_o(t)$ denota que as condições iniciais são nulas (resposta ao estado nulo)

Condições Iniciais Nulas

$X(s) = 1/s$ → $\boldsymbol{H(s) = \dfrac{6.25}{s + 6.25}}$ → $Y(s) = X(s)H(s) = \dfrac{1}{s}H(s)$

Resposta ao degrau

Como $x(t) = u(t) \xrightarrow{\mathcal{L}} X(s) = 1/s$

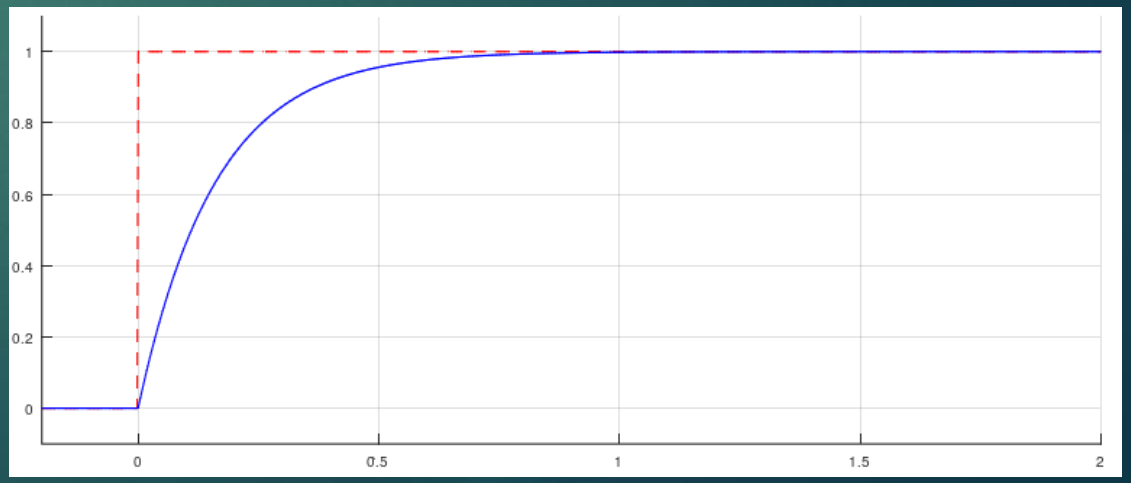
$$Y_o(s) = X(s)H(s) = \left(\frac{1}{s}\right)H(s) = \frac{6.25}{s(s+6.25)}$$

$$Y_o(s) = X(s)H(s) = \frac{6.25}{s(s+6.25)} = \frac{A}{s} + \frac{\mathrm{B}}{(s+6.25)} = \frac{1}{s} + \frac{-1}{(s+6.25)}$$

$$y_o(t) = 1 - e^{-6.25t}$$

```
>> Hs=6.25/(s+6.25);
>> Xs=1/s;
>> ft=ilaplace_fga(Hs*Xs,false);
ft=@(t)(+(-1.00000*exp(-6.25000*t))+(1.00000) ).*f_u(t);
```

```
>> RespostaSistemaHs(Hs,1/s,0,1,true)
--Saida--
ft=@(t)(+(-1.00000*exp(-6.25000*t))+(1.00000) ).*f_u(t);

--Entrada--
fx=@(t)(y=(1.00000)*f_u(t)+(0.00000)*f_d(t));
```



Calcular $y(t)$ se $x(t) = 10.5e^{-t}$ Resposta ao estado nulo

O índice '0' em $Y_o(s)$ e $y_o(t)$ denota que as condições iniciais são nulas (resposta ao estado nulo)

Condições Iniciais
$y(0^-) = 0$

$$X(s) = \frac{10.5}{(s+1)} \longrightarrow \boxed{\boldsymbol{H(s) = \frac{6.25}{s+6.25}}} \longrightarrow Y_o(s) = X(s)H(s)$$

Como $x(t) = 10.5e^{-t} \xrightarrow{\mathcal{L}} X(s) = \frac{10.5}{(s+1)}$

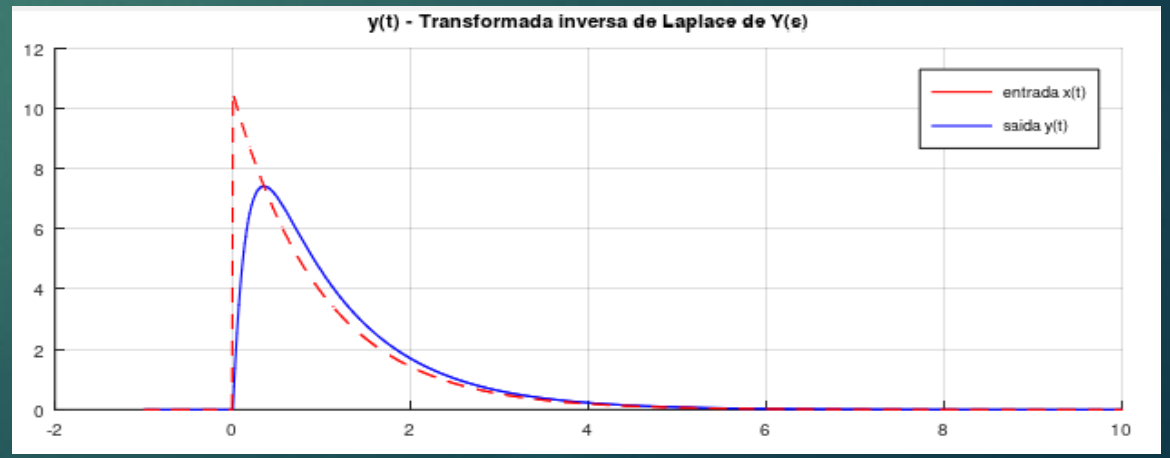
$$Y_o(s) = X(s)H(s) = \frac{10.5}{(s+1)}\frac{6.25}{(s+6.25)}$$

$$Y_o(s) = \frac{65.625}{(s+1)(s+6.25)} = \frac{\mathrm{A}}{(s+1)} + \frac{\mathrm{B}}{(s+6.25)} = \frac{12.5}{(s+1)} + \frac{-12.5}{(s+6.25)}$$

$$Y_o(s) \xRightarrow{\mathcal{L}^{-1}} y_o(t) = [12.5e^{-t} - 12.5e^{-6.25t}]\mathrm{u(t)}$$

```
>> Hs=6.25/(s+6.25);
>> Xs=10.5/(s+1);
>> ft=ilaplace_fga(Hs*Xs,false);
ft=@(t)(+(-12.50000*exp(-6.25000*t))+(12.50000*exp(-1.00000*t)) ).*f_u(t);
```

```
>> RespostaSistemaHs(Hs,10.5/(s+1),0,1,true);
--Saida--
ft=@(t)(+(-12.50000*exp(-6.25000*t))+(12.50000*exp(-1.00000*t)) ).*f_u(t);

--Entrada--

ft=@(t)(+(10.50000*exp(-1.00000*t)) ).*f_u(t);
```



Recalcular $y(t)$ *se* $x(t) = 10.5e^{-t}$ *adicioando condições iniciais*

$$Y(s) = Y_o(s) + X_n(s)\left(\frac{1}{H(s).den}\right)$$

$$Y_o(s) = X(s)H(s) = \frac{10.5}{(s+1)}\frac{6.25}{(s+6.25)}$$

$$y_o(t) = [12.5e^{-t} - 12.5e^{-6.25t}]\mathrm{u(t)}$$

Como $y(0) = 5 \rightarrow X_n(s) = 5$

$$Y(s) = Y_o(s) + (5)\left(\frac{1}{(s+6.25)}\right)$$

$$(5)\left(\frac{1}{(s+6.25)}\right) \xrightarrow{\mathcal{L}^{-1}} y_n(t) = 5e^{-6.25t}$$

$$y(t) = y_o(t) + y_n(t) = [12.5e^{-t} - 12.5e^{-6.25t}]\mathrm{u(t)} + [5e^{-6.25t}]\mathrm{u(t)}$$

$$y(t) = (12.5e^{-t} - 7.5e^{-6.25t})\mathrm{u(t)}$$

Condições Iniciais
$y(0^-) = 5$

$$X(s) = \frac{10.5}{(s+1)}$$

$$X_n(s) = 5$$

$$\boldsymbol{H(s) = \frac{6.25}{s+6.25}}$$

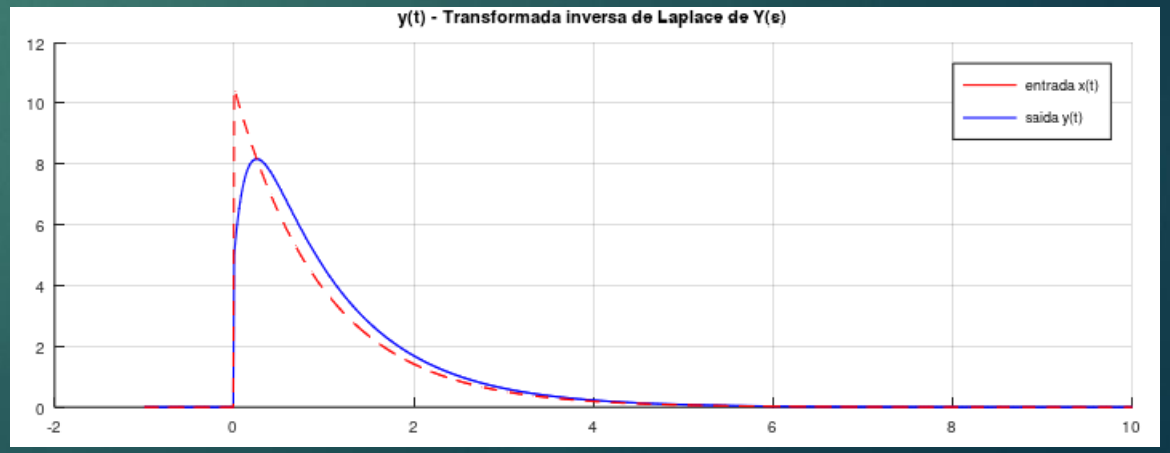
$$Y(s) = Y_o(s) + Y_n(s)$$

$$\boldsymbol{H_n(s) = \frac{1}{H(s).den} = \frac{1}{s+a}}$$

$$X_n(s) = y(0)$$

```
>> Hs=6.25/(s+6.25);
>> Xs=10.5/(s+1);
>> [num_c,den_c]=tfdata(Hs,'v');
>> Hn=tf(1,den_c);
>> Xn=5;
>> ft=ilaplace_fga(Hs*Xs+Hn*Xn,false);
ft=@(t)(+(-7.50000*exp(-6.25000*t))+(12.50000*exp(-1.00000*t)) ).*f_u(t);
```

```
>> RespostaSistemaHs(Hs,10.5/(s+1),5,1,true);
--Saida--
ft=@(t)(+(-7.50000*exp(-6.25000*t))+(12.50000*exp(-1.00000*t)) ).*f_u(t);

--Entrada--

ft=@(t)(+(10.50000*exp(-1.00000*t)) ).*f_u(t);
```



Calcular $y(t)$ se $x(t) = x(t) = 10\cos(7t)u(t)$ Resposta ao estado nulo

Como $x(t) = 10\cos(7t)u(t) \xrightarrow{\mathcal{L}} X(s) = \frac{10s}{(s^2+7^2)}$

$$Y_o(s) = X(s)H(s) = \frac{10s}{(s^2+7^2)}\frac{6.25}{(s+6.25)}$$

$$Y_o(s) = \frac{62.5}{(s^2+7^2)(s+6.25)} = \frac{A}{s+6.25} + \frac{Cs+D}{s^2+7^2} = \frac{-4.436}{(s+6.25)} + \frac{4.436s+34.78}{s^2+7^2}$$

$$Y_o(s) \xRightarrow{\mathcal{L}^{-1}} y_o(t) = (4.436\cos(7t) + 4.969\sin(7t) - 4.4366e^{-6.25t})u(t)$$

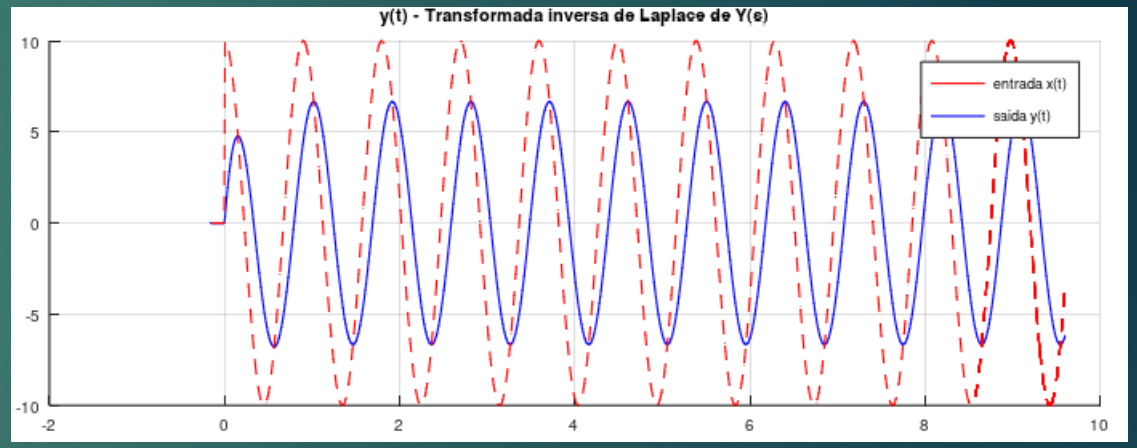
Condições Iniciais

$y(0^-) = 0$

$y'(0^-) = 0$

$$X(s) = \frac{10s}{(s^2+7^2)} \Rightarrow \boxed{H(s) = \frac{6.25}{s+6.25}} \Rightarrow Y_o(s) = X(s)H(s)$$

```
>> Hs=6.25/(s+6.25);
>> Xs=10*s/(s^2+7^2);
>> ft=ilaplace_fga(Hs*Xs,true);
ft=@(t)(+(2*(3.33008)*cos(7.00000*t+(-0.84194)))+
(-4.43577*exp(-6.25000*t)) ).*f_u(t);
```




```
--Entrada--
ft=@(t)(+(2*(5.00000)*cos(7.00000*t+(-0.00000))) ).*f_u(t);
>> RespostaSistemaHs(Hs,10*s/(s^2+7^2),0,6,true);
--Saida--
ft=@(t)(+(2*(3.33008)*cos(7.00000*t+(-0.84194)))+(-4.43577*exp(-6.25000*t)) ).*f_u(t);

--Entrada--
ft=@(t)(+(2*(5.00000)*cos(7.00000*t+(-0.00000))) ).*f_u(t);
```

Recalcular $y(t)$ se $x(t) = x(t) = 10\cos(7t)u(t)$ com condições iniciais

$$Y(s) = Y_o(s) + X_n(s)\left(\frac{1}{H(s).den}\right)$$

$$Y_o(s) = X(s)H(s) = \frac{10s}{(s^2+7^2)}\frac{6.25}{(s+6.25)}$$

$$y_o(t) = (4.436\cos(7t) + 4.969\sin(7t) - 4.436e^{-6.25t})\mathrm{u(t)}$$

Como $y(0) = 5 \rightarrow X_n(s) = 5$

$$Y(s) = Y_o(s) + (5)\left(\frac{1}{(s+6.25)}\right)$$

$$(5)\left(\frac{1}{(s+6.25)}\right) \xrightarrow{\mathcal{L}^{-1}} y_n(t) = 5e^{-6.25t}$$

$$y(t) = y_o(t) + y_n(t) = (4.436\cos(7t) + 4.969\sin(7t) - 4.436e^{-6.25t}) + [5e^{-6.25t}]\mathrm{u(t)}$$

$$y(t) = (4.436\cos(7t) + 4.969\sin(7t) + 0.564e^{-6.25t})\mathrm{u(t)}$$

$$X(s) = \frac{10s}{(s^2+7^2)}$$

Condições Iniciais
$y(0^-) = 5$

$$H(s) = \frac{6.25}{s+6.25}$$

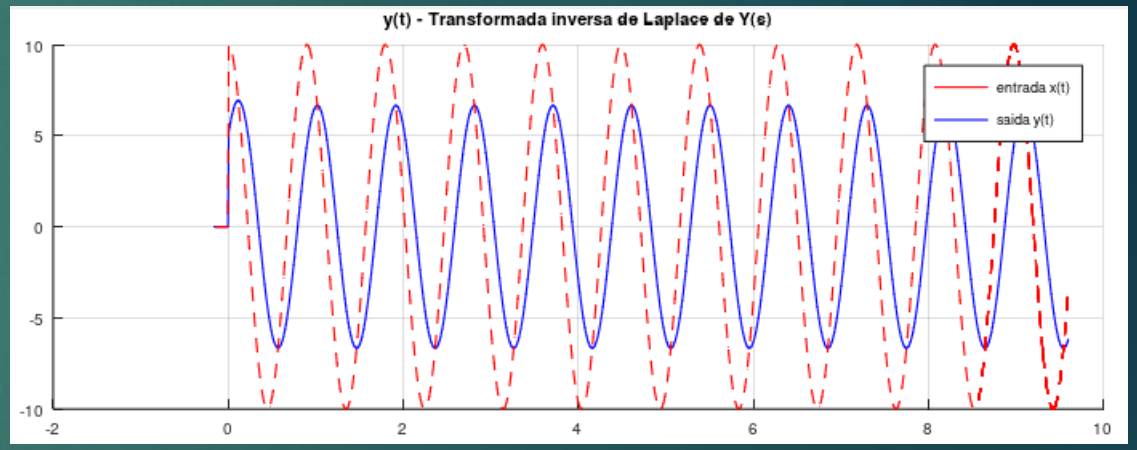
$$Y(s) = Y_o(s) + Y_n(s)$$

$$X_n(s) = 5$$

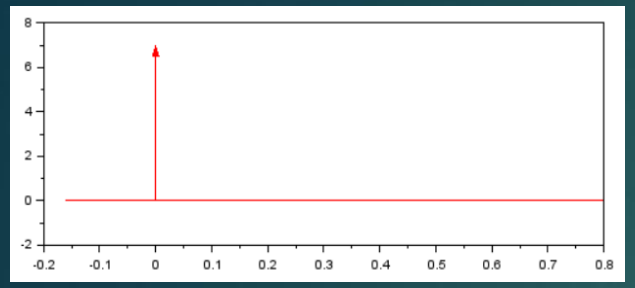
$$H_n(s) = \frac{1}{H(s).den} = \frac{1}{s+a}$$

$$X_n(s) = y(0)$$

```
>> Hs=6.25/(s+6.25);
>> Xs=10*s/(s^2+7^2);
>> [num_c,den_c]=tfdata(Hs,'v');
>> Hn=tf(1,den_c);
>> Xn=5;
>> Ys = soma_tf(Hs*Xs,Hn*Xn);
>> ft=ilaplace_fga(Ys,false);
ft=@(t)(+(2*(2.21789*cos(7.00000*t)-(-2.48403)*sin(7.00000*t)))+
(0.56423*exp(-6.25000*t)) ).*f_u(t);
```




```
>> RespostaSistemaHs(Hs,10*s/(s^2+7^2),5,6,true);
--Saida--
ft=@(t)(+(2*(3.33008)*cos(7.00000*t+(-0.84194)))+(0.56423*exp(-6.25000*t)) ).*f_u(t);

--Entrada--
ft=@(t)(+(2*(5.00000)*cos(7.00000*t+(-0.00000))) ).*f_u(t);
```



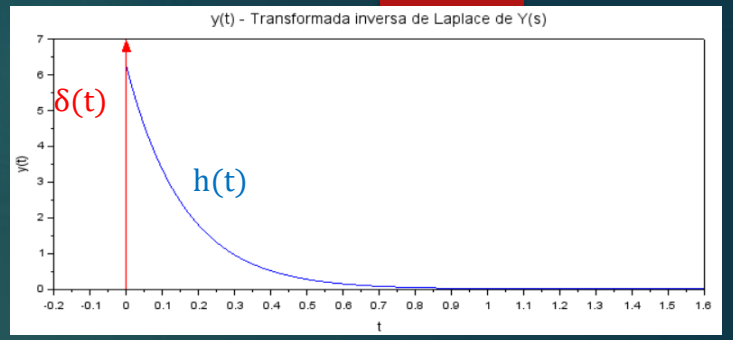
$x(t) = \delta(t)$

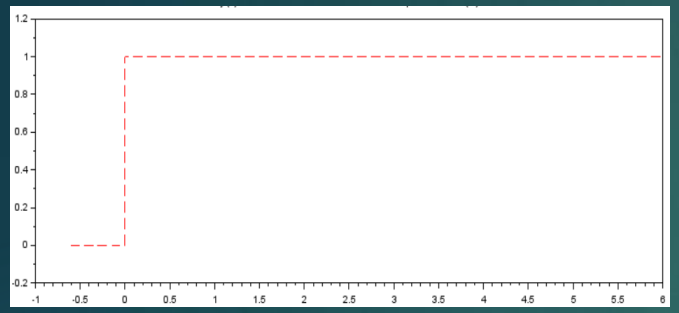
$X(s) = 1$

$$\boldsymbol{H(s)} = \frac{6.25}{(s + 6.25)}$$

$y(0^-) = 0$

$h(t) = 6.25\, e^{-6.25t}$

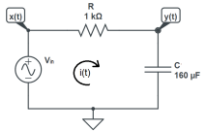




$x(t) = u(t)$

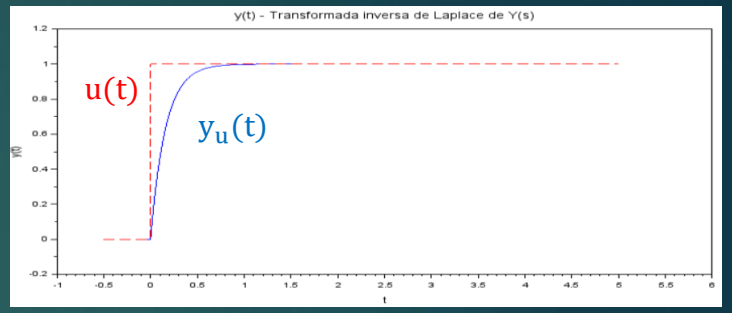
$X(s) = \frac{1}{s}$

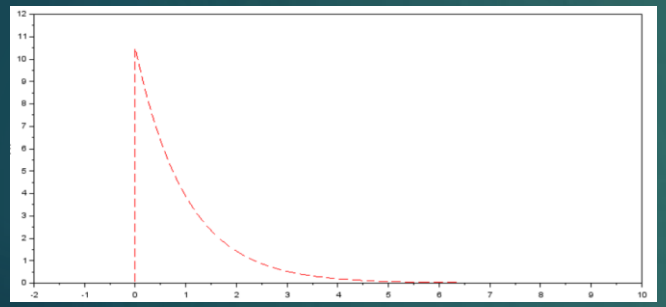
$$\boldsymbol{H(s)} = \frac{6.25}{(s + 6.25)}$$



$y(0^-) = 0$

$y_u(t) = 1 - e^{-6.25t}$





$x(t) = 10.5\, e^{-t}$

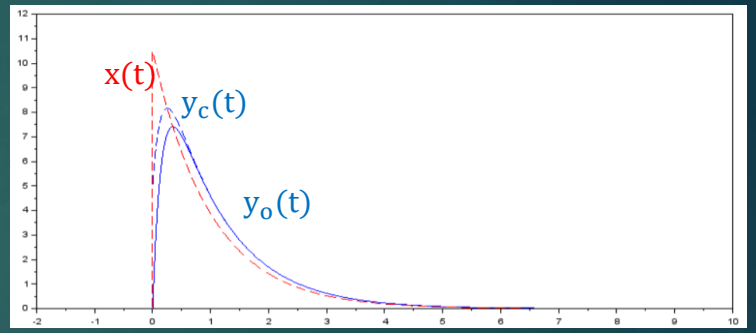
$X(s) = \frac{10.5}{(s + 1)}$

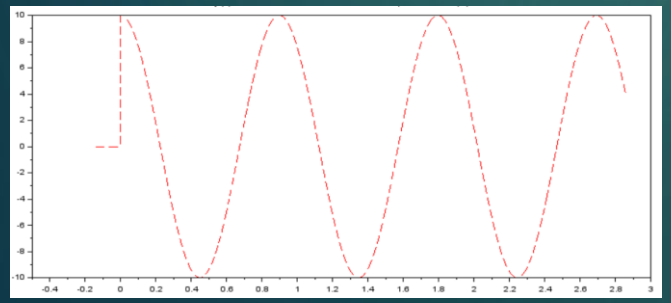
$$\boldsymbol{H(s)} = \frac{6.25}{(s + 6.25)}$$

$y(0^-) = 5$

$y_o(t) = -12.5\, e^{-6.25t} + 12.5e^{-t}$

$y_c(t) = -7.5e^{-6.25t} + 12.5e^{-t}$





$x(t) = 10\cos(7t)$

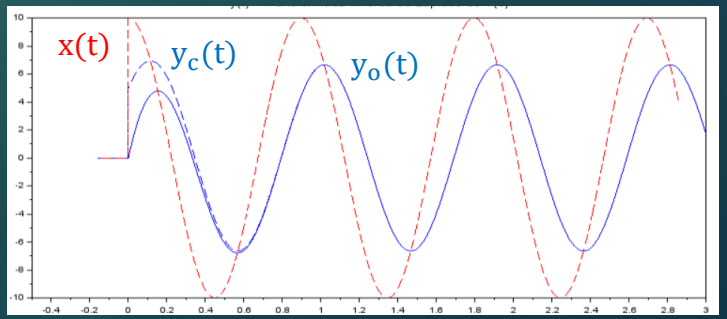
$X(s) = \frac{10s}{(s^2 + 7^2)}$

$$\boldsymbol{H(s)} = \frac{6.25}{(s + 6.25)}$$

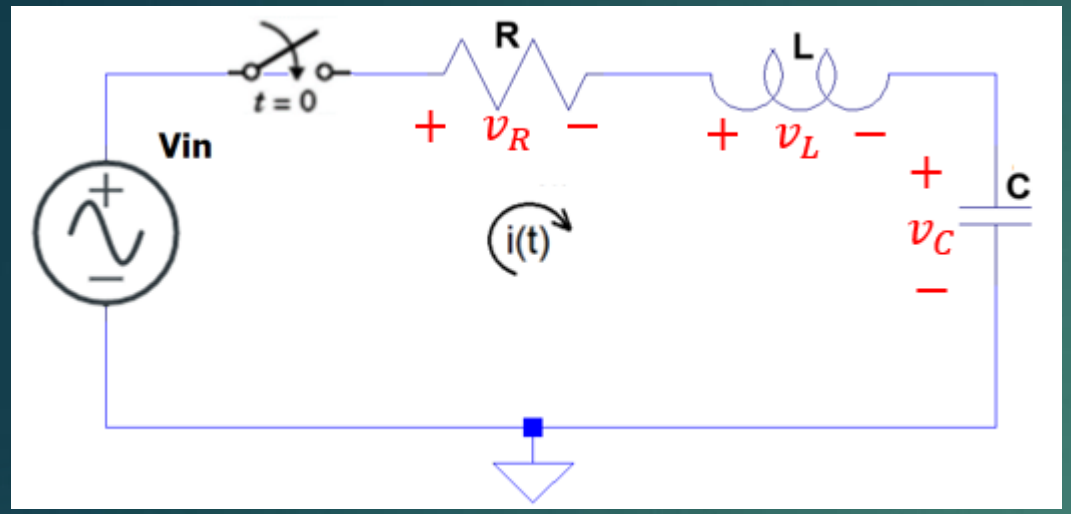
$y(0^-) = 5$

$y_o(t) = -\,4.44e^{-6.25t} + 6.66\cos(7t - 0.842)$

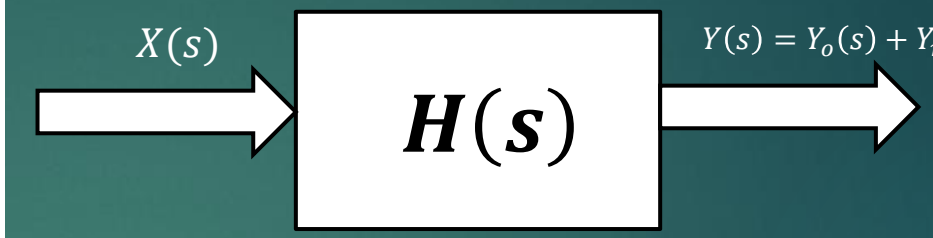
$y_c(t) = 0.56\, e^{-6.25t} + 6.66\cos(7t - 0.842)$



Sistema Elétrico 2ª Ordem



Condições Iniciais
$y(0^-) = 20V$
$y'(0^-) = 0$



$$Y(s) = Y_o(s) + Y_n(s) = X(s)H(s) + X_n\left(\frac{1}{H(s).den}\right)$$

$x(t)$ – fonte de tensão

$y(t)$ – corrente no capacitor

Este sistema elétrico é definido pela seguinte equação diferencial:

$$v_c''(\mathrm{t}) + \frac{\mathrm{R}}{\mathrm{L}}v_c'(t) + \frac{1}{\mathrm{LC}}v_c(t) = \frac{1}{LC}v_{in}(t)$$

$$y''(t) + 120y'(t) + 2700y(t) = 2700x(t)$$

A entrada é representada pelo sinal $v_{in}(\mathrm{t})$ e a saída pelo sinal $v_c(\mathrm{t})$

Condições Iniciais
$y(0^-) = 20V$
$y'(0^-) = 0$

$R = 12\Omega$

$L = 0.1H$

$C = 3.7mF$

Função de Transferência do Sistema

A função de transferência no domínio de Laplace é definida como sendo a razão entre a transformada de Laplace do sinal de saída pela transformada de Laplace do sinal de entrada

$$H(s) = \frac{Y(s)}{X(s)} = \frac{2700}{(s^2 + 120s + 2700)} = \frac{27700}{(s+90)(s+30)}$$

Sem Condições Iniciais

$y''(t) + 120y'(t) + 2700y(t) = 2700x(t)$

Aplicacando Laplace em ambos os lados

$s^2Y_o(s) + 120sY_o(s) + 2700Y_o(s) = 2700X(s)$

$(s^2 + 120s + 2700)Y_o(s) = 2700X(s)$

$$H(s) = \frac{Y_o(s)}{X(s)} = \frac{2700}{(s^2 + 120s + 2700)}$$

$Y_o(s) = X(s)H(s)$

Entrada $x(t) \xrightarrow{\mathcal{L}} X(s)$

Saída $y_o(t) \xrightarrow{\mathcal{L}} Y_o(s)$

$y'(t) \xrightarrow{\mathcal{L}} sY_o(s)$

$y(0^-) = 0$

$y'(0^-) = 0$

$X(s) \rightarrow \boxed{\boldsymbol{H(s)} = \frac{2700}{(s+90)(s+30)}} \rightarrow Y_o(s) = X(s)H(s)$

$$\boldsymbol{H_n(s)} = \frac{\mathbf{1}}{\boldsymbol{H(s).den}} = \frac{\mathbf{1}}{\boldsymbol{s^2 + as + b}}$$

$X_n(s) = y(0)(s + a) + y'(0)$

Com Condições Iniciais

$s^2Y(s) - sy(0) - y'(0) + 120\big(sY(s) - y(0)\big) + 2700Y(s) = 2700X(s)$

$(s^2 + 120s + 2700)Y(s) - sy(0) - y'(0) - 120y(0) = 6X(s)$

$$Y(s) = X(s)\frac{2700}{(s^2 + 120s + 2700)} + \big(sy(0) + y'(0) + 120y(0)\big)\frac{1}{(s^2 + 120s + 2700)}$$

$X_n(s) = sy(0) + y'(0) + 120y(0) = y(0)(s + a) + y'(0)$

$$Y(s) = X(s)H(s) + X_n(s)\left(\frac{1}{H(s).den}\right)$$

$$Y(s) = Y_o(s) + X_n(s)\left(\frac{1}{H(s).den}\right)$$

$X(s) \rightarrow \boxed{\boldsymbol{H(s)} = \frac{2700}{(s+90)(s+30)}} \rightarrow Y_o(s) = X(s)H(s)$

$y(0^-) = 20V$

$y'(0^-) = 0$

$X_n(s) = 20(s + 120) \rightarrow \boxed{\boldsymbol{H_n(s)} = \frac{1}{(s+90)(s+30)}} \rightarrow Y_n(s) = X_n(s)\frac{1}{H(s).den}$

Calcular a Resposta ao impulso h(t) do Sistema: $x(t) = \delta(t)$

Condições Iniciais Nulas

$x(t) = \delta(t)$

resposta ao estado nulo

$X(s) = 1$

Condições Iniciais Nulas

$$\boldsymbol{H(s)} = \frac{2700}{(s+90)(s+30)}$$

$Y(s) = X(s)H(s) = H(s)$

Resposta ao impulso

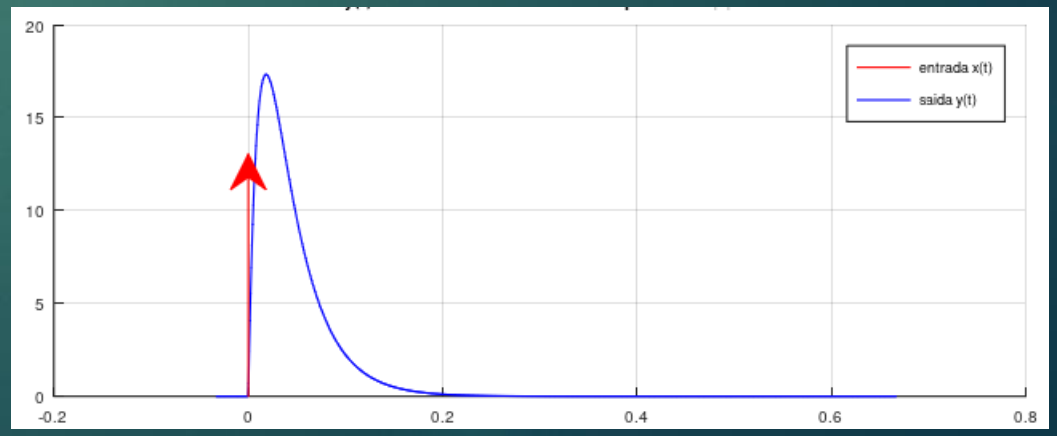
Como $x(t) = \delta(t) \xrightarrow{\mathcal{L}} X(s) = 1$ então $Y(s) = X(s)H(s) = (1)H(s) = \dfrac{2700}{(s^2+120s+2700)}$

Então a resposta $h(t)$ do sistema ao impulso será a transformada inversa de Laplace da própria função de transferência

$$H(s) = \frac{2700}{(s+90)(s+30)} = \frac{\mathrm{A}}{(s+30)} + \frac{\mathrm{B}}{(s+90)} = \frac{45}{(s+30)} + \frac{-45}{(s+90)} \xrightarrow{\mathcal{L}^{-1}} h(t) = [45e^{-30t} - 45e^{-90t}]\mathrm{u(t)}$$

```
>> Hs=(2700)/(s^2+120*s+2700);
>> Xs=1;
>> ft=ilaplace_fga(Hs*Xs,true);
ft=@(t)(+(-45.00000*exp(-90.00000*t))+(45.00000*exp(-30.00000*t)) ).*f_u(t);
```

```
>> RespostaSistemaHs(Hs,1,[0 0],1,true);
--Saida--
ft=@(t)(+(-45.00000*exp(-90.00000*t))+(45.00000*exp(-30.00000*t)) ).*f_u(t);

--Entrada--
fx=@(t)(y=(1.00000)*f_d(t));
```



Calcular a Resposta ao Degrau y(t) do Sistema: $x(t) = u(t)$

Condições Iniciais *nulas*

resposta ao estado nulo

$x(t) = u(t)$

$X(s) = 1/s$

Condições Iniciais Nulas

$$\boldsymbol{H(s)} = \frac{2700}{(s+90)(s+30)}$$

Resposta ao degrau

$$Y(s) = X(s)H(s) = \frac{1}{s}H(s)$$

Como $x(t) = u(t) \xrightarrow{\mathcal{L}} X(s) = 1/s$ então $Y(s) = X(s)H(s) = \left(\frac{1}{s}\right)H(s) = \frac{1}{S}\frac{2700}{(s+90)(s+30)}$

$$Y(s) = X(s)H(s) = \frac{2700}{s(s+90)(s+30)} = \frac{A}{s} + \frac{B}{(s+30)} + \frac{C}{(s+90)} = \frac{1}{s} + \frac{-1.5}{(s+30)} + \frac{0.5}{(s+90)} \xrightarrow{\mathcal{L}^{-1}} y(t) = [1 - 1.5e^{-30t} + 0.5e^{-90t}]u(t)$$

```
>> Hs=(2700)/(s^2+120*s+2700);
>> Xs=1/s;
>> ft=ilaplace_fga(Hs*Xs,false);
ft=@(t)(+(0.50000*exp(-90.00000*t))+(-1.50000*exp(-30.00000*t))+(1.00000) ).*f_u(t);
```

```
>> RespostaSistemaHs(Hs,1/s,[0 0],1,true);
--Saida--
ft=@(t)(+(0.50000*exp(-90.00000*t))+(-1.50000*exp(-30.00000*t))+(1.00000) ).*f_u(t);

--Entrada--
fx=@(t)(y=(1.00000)*f_u(t)+(0.00000)*f_d(t));
```

Calcular $y(t)$ se $x(t) = 10e^{-15t}$ Resposta ao estado nulo

O índice '0' em $Y_o(s)$ e $y_o(t)$ denota que as condições iniciais são nulas (resposta ao estado nulo)

Condições Iniciais
$y(0^-) = 0$
$y'(0^-) = 0$

$$X(s) = \frac{10}{(s+15)} \longrightarrow \boxed{H(s) = \frac{2700}{(s+90)(s+30)}} \longrightarrow Y_o(s) = X(s)H(s)$$

Como $x(t) = 10e^{-15t} \xrightarrow{\mathcal{L}} X(s) = \frac{10}{(s+15)}$ então $Y_o(s) = X(s)H(s) = \frac{10}{(s+15)} \frac{2700}{(s+30)(s+90)}$

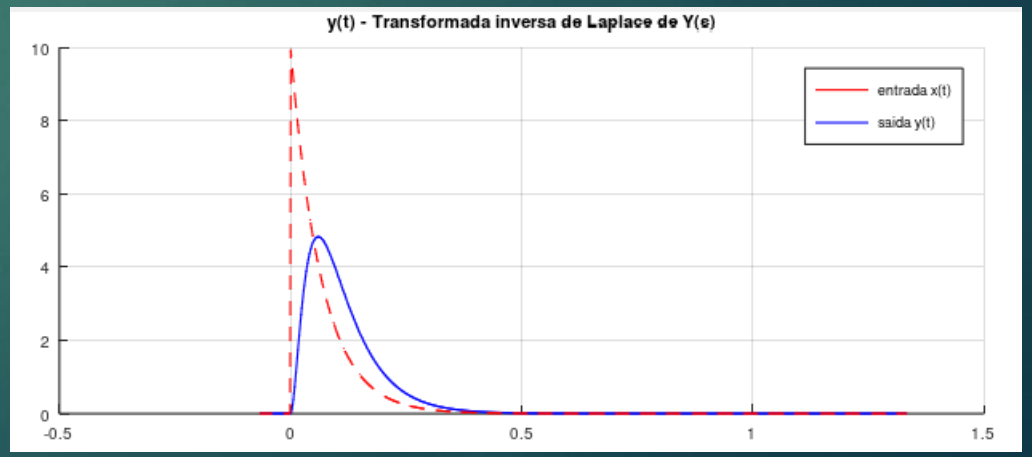
$$Y_o(s) = \frac{27000}{(s+15)(s+30)(s+90)} = \frac{A}{(s+15)} + \frac{B}{(s+30)} + \frac{C}{(s+90)} = \frac{24}{(s+15)} + \frac{-30}{(s+30)} + \frac{6}{(s+90)}$$

$$Y_o(s) \xRightarrow{\mathcal{L}^{-1}} y_o(t) = (24e^{-15t} - 30e^{-30t} + 6e^{-90t})\mathrm{u(t)}$$

$$y_o(t) = (24e^{-15t} - 30e^{-30t} + 6e^{-90t})\mathrm{u(t)}$$

```
>> Hs=(2700)/(s^2+120*s+2700);
>> Xs=10/(s+15);
>> ft=ilaplace_fga(Hs*Xs,false);
ft=@(t)(+(6.00000*exp(-90.00000*t))+(-30.00000*exp(-30.00000*t))+
(24.00000*exp(-15.00000*t)) ).*f_u(t);
```

```
>> RespostaSistemaHs(Hs,10/(s+15),[0 0],1,true);
--Saida--
ft=@(t)(+(6.00000*exp(-90.00000*t))+(-30.00000*exp(-30.00000*t))+
(24.00000*exp(-15.00000*t)) ).*f_u(t);

--Entrada--

ft=@(t)(+(10.00000*exp(-15.00000*t)) ).*f_u(t);
```



Recalcular $y(t)$ se $x(t) = 10e^{-15t}$ adicioando condições iniciais $y(0) = 5$ e $y'(0) = 0$

$$Y(s) = Y_o(s) + X_n(s)\left(\frac{1}{H(s).den}\right)$$

$$Y_o(s) = X(s)H(s) = \frac{10}{(s+15)}\frac{2700}{(s+90)(s+30)}$$

$$y_o(t) = (24e^{-15t} - 30e^{-30t} + 6e^{-90t})\mathrm{u(t)}$$

Como $y(0) = 20$ e $y'(0) = 0$ $\rightarrow X_n(s) = 5s + 600$

$$Y(s) = Y_o(s) + (5s + 600)\left(\frac{1}{(s+90)(s+30)}\right)$$

$$5(s+120)\left(\frac{1}{(s+90)(s+30)}\right) \xrightarrow{\mathcal{L}^{-1}} y_n(t) = 7.5e^{-30t} - 2.5e^{-90t}$$

$$y_c(t) = y_o(t) + y_n(t) = (24e^{-15t} - 30e^{-30t} + 6e^{-90t})\mathrm{u(t)} + [7.5e^{-30t} - 2.5e^{-90t}]\mathrm{u(t)}$$

$$y_c(t) = 24e^{-15t} - 22.5e^{-30t} + 3.5e^{-90t}$$

```
>> Hs=(2700)/(s^2+120*s+2700);
>> Xs=10/(s+15);
>> [num_c,den_c]=tfdata(Hs,'v');
>> Hn=tf(1,den_c);
>> Xn=5*s+600;
>> ft=ilaplace_fga(Hs*Xs+Hn*Xn,false);
ft=@(t)(+(3.50000*exp(-90.00000*t))+(-22.50000*exp(-30.00000*t))+
(24.00000*exp(-15.00000*t)) ).*f_u(t);
```

```
>> RespostaSistemaHs(Hs,10/(s+15),[5 0],1,true);
--Saida--
ft=@(t)(+(3.50000*exp(-90.00000*t))+(-22.50000*exp(-30.00000*t))+
(24.00000*exp(-15.00000*t)) ).*f_u(t);

--Entrada--
ft=@(t)(+(10.00000*exp(-15.00000*t)) ).*f_u(t);
```

$$X(s) = \frac{10}{(s+15)}$$

$$X_n(s) = 5(s+120)$$

Condições Iniciais

$y(0^-) = 5$

$y'(0^-) = 0$

$$H(s) = \frac{2700}{(s+90)(s+30)}$$

$$Y(s) = Y_o(s) + X_n(s)\left(\frac{1}{H(s).den}\right)$$

$$H_n(s) = \frac{1}{H(s).den} = \frac{1}{s^2 + as + b}$$

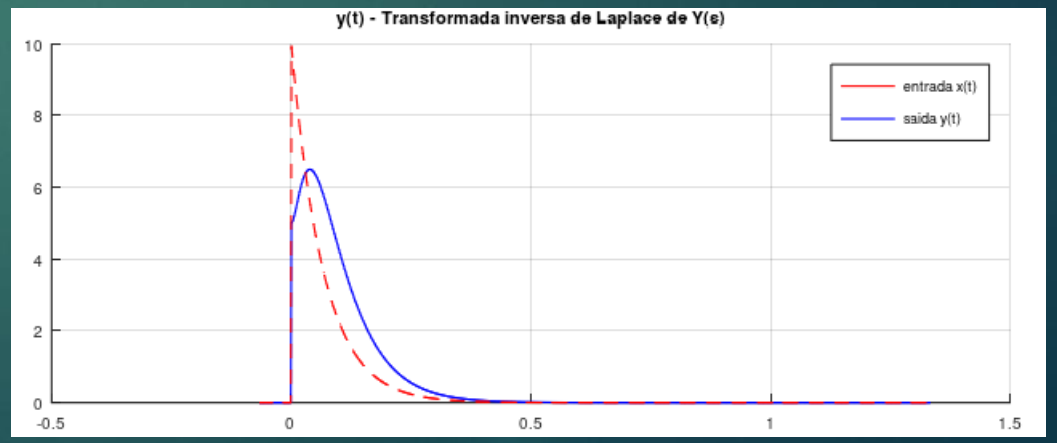
$$X_n(s) = y(0)(s+a) + y'(0)$$

$$H(s) = \frac{Y_o(s)}{X(s)} = \frac{2700}{(s^2 + 120s + 2700)}$$

$a = 120;\quad y(0) = 5;\quad y'(0) = 0$

```
>> Hs=(2700)/(s^2+120*s+2700);
>> Xs=EntradaNula(Hs,[5,0])

Transfer function 'Xs' from input

 y1:  5 s + 600
```

```
function  Xs=EntradaNula(Hs,y_ini)
    [num_c,den_c]=tfdata(Hs,'v');
    num_c=flip(num_c);
    den_c = flip(den_c);
    ny = length(y_ini);
    ncond=length(den_c)-1;
    y=y_ini;
    if ny < ncond
    ncond=length(den_c)-1;
    if length(y) < length(den_c)-1
    p = den_c;
    Xs=0;
    s=tf('s');
    for cond=1:length(y)
endfunction
```



Calcular $y(t)$ se $x(t) = 37\cos(15t)$ Resposta ao estado nulo

Condições Iniciais
$y(0^-) = 0$
$y'(0^-) = 0$

$$X(s) = \frac{37s}{(s^2+15^2)} \Rightarrow \boxed{H(s) = \frac{2700}{(s+90)(s+30)}} \Rightarrow Y_o(s) = X(s)H(s)$$

Como $x(t) = 37\cos(15t) \xrightarrow{\mathcal{L}} X(s) = \frac{37s}{(s^2+15^2)}$

então $Y_o(s) = X(s)H(s) = \frac{37s}{(s^2+15^2)}\frac{2700}{(s+90)(s+30)}$

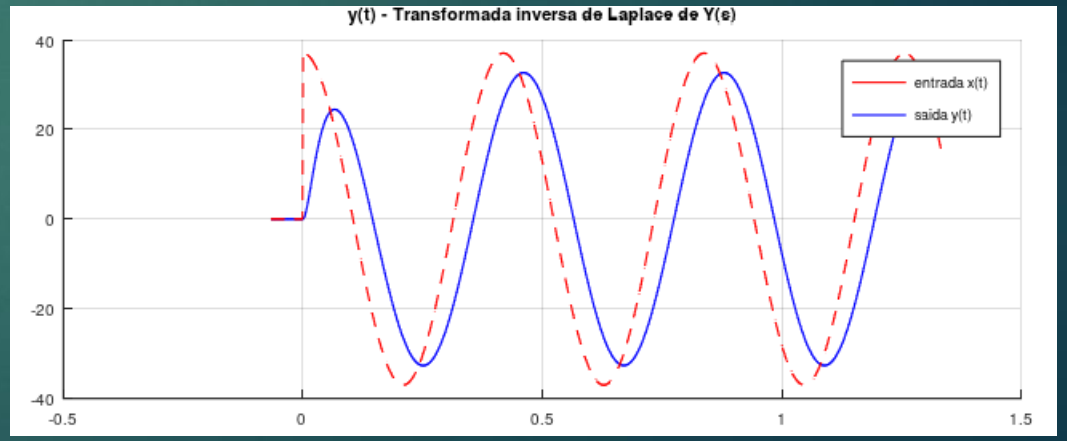
$$Y_o(s) = \frac{(37)(2700)s}{(s^2+15^2)(s+90)(s+30)} \xrightarrow{\mathcal{L}^{-1}} y_o(t)$$

$$y_o(t) = (-44.4e^{-30t} + 18e^{-90t} + 26.4\cos(15t) + 19.2\sin(15t))u(t)$$

O índice '0' em $Y_o(s)$ e $y_o(t)$ denota que as condições iniciais são nulas(**RESPOSTA AO ESTADO NULO**).

```
>> Hs=(2700)/(s^2+120*s+2700);
>> Xs=37*s/(s^2+15^2);
>> ft=ilaplace_fga(Hs*Xs,false);
ft=@(t)(+(18.00000*exp(-90.00000*t))+(-44.40000*exp(-30.00000*t))+(
2*(13.20000*cos(15.00000*t)-(-9.60000)*sin(15.00000*t))) ).*f_u(t);
```

```
>> RespostaSistemaHs(Hs,37*s/(s^2+15^2),[0 0],1,false);
--Saida--
ft=@(t)(+(18.00000*exp(-90.00000*t))+(-44.40000*exp(-30.00000*t))+(
2*(13.20000*cos(15.00000*t)-(-9.60000)*sin(15.00000*t))) ).*f_u(t);

--Entrada--
ft=@(t)(+(2*(18.50000*cos(15.00000*t)-(-0.00000)*sin(15.00000*t)))
).*f_u(t);
```



Recalcular $y(t)$ se $x(t) = 37\cos(15t)$ Com condições Iniciais $y(0) = 20$ e $y'(0) = 0$

$$Y(s) = X(s)H(s) + X_n(s)\left(\frac{1}{H(s).den}\right)$$

$$Y_o(s) = X(s)H(s) = \frac{37s}{(s^2+15^2)}\frac{2700}{(s+90)(s+30)}$$

$$y_o(t)=(-44.4e^{-30t} + 18e^{-90t} + 26.4cos(15t) + 19.2\,sin(15t))u(t)$$

Como $y(0) = 5$ e $y'(0) = 0$ $\rightarrow X_n(s) = 5s + 600$

$$Y(s) = Y_o(s) + 20(s+120)\left(\frac{1}{(s+90)(s+30)}\right)$$

$$20(s+120)\left(\frac{1}{(s+90)(s+30)}\right) \xrightarrow{\mathcal{L}^{-1}} y_n(t) = 30e^{-30t} - 10e^{-90t}$$

$$y(t) = y_o(t) + y_n(t)$$
$$= (-44.4e^{-30t} + 18e^{-90t} + 26.4cos(15t) + 19.2\,sin(15t)) + (30e^{-30t} - 10e^{-90t})$$

$$y(t) = (-14.4e^{-30t} + 8.0e^{-90t} + 26.4cos(15t) + 19.2\,sin(15t))u(t)$$

$$X(s) = \frac{37s}{(s^2+15^2)}$$

Condições Iniciais
$y(0^-) = 20$
$y'(0^-) = 0$

$$\boxed{H(s) = \frac{2700}{(s+90)(s+30)}}$$

$$X_n(s) = 20(s+120)$$

$$Y(s) = X(s)H(s) + X_n(s)\left(\frac{1}{H(s).den}\right)$$

$$\boxed{\begin{aligned} H_n(s) &= \frac{1}{H(s).den} = \frac{1}{s^2+as+b} \\ X_n(s) &= y(0)(s+a) + y'(0) \end{aligned}}$$

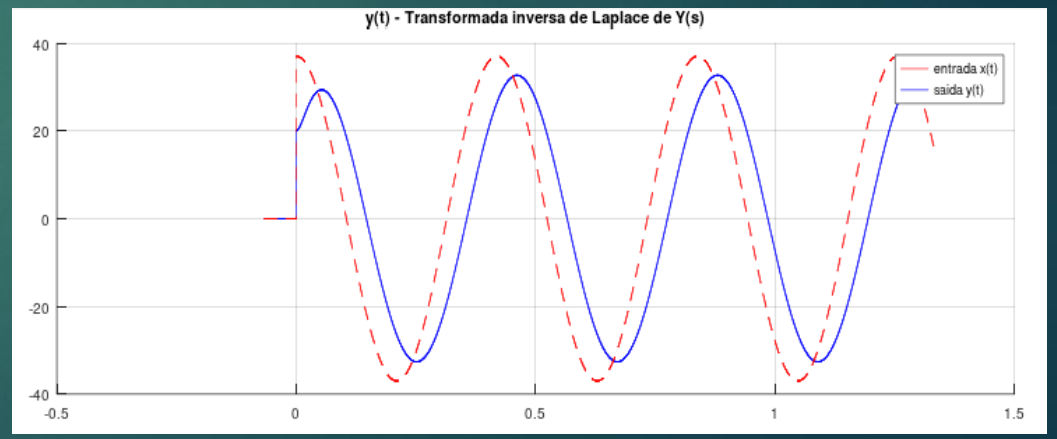
$a = 120;\quad y(0) = 20\quad y'(0) = 0$

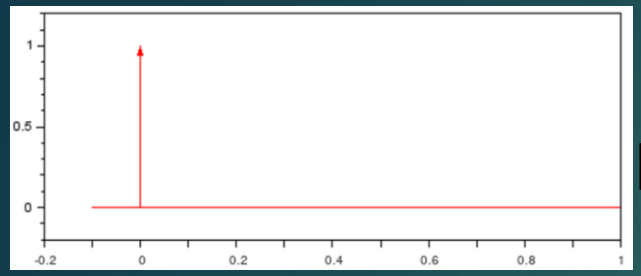
$$X_n(s) = 20(s+120)$$

```
>> Hs=(2700)/(s^2+120*s+2700);
>> Xn=EntradaNula(Hs,[20,0])

Transfer function 'Xn' from inp

 y1:  20 s + 2400
```

```
>> Hs=(2700)/(s^2+120*s+2700);
>> Xs=37*s/(s^2+15^2);
>> [num_c,den_c]=tfdata(Hs,'v');
>> Hn=tf(1,den_c);
>> Xn=20*s+2400;
>> ft=ilaplace_fga(Hs*Xs+Hn*Xn,false);
ft=@(t)(+(8.00000*exp(-90.00000*t))+(-14.40000*exp(-30.00000*t))+(2
*(13.20000*cos(15.00000*t)-(-9.60000)*sin(15.00000*t))) ).*f_u(t);
```

```
>> RespostaSistemaHs(Hs,37*s/(s^2+15^2),[20 0],1,false);
--Saida--
ft=@(t)(+(8.00000*exp(-90.00000*t))+(-14.40000*exp(-30.00000*t))+(2*(13.20000
*cos(15.00000*t)-(-9.60000)*sin(15.00000*t))) ).*f_u(t);

--Entrada--
ft=@(t)(+(2*(18.50000*cos(15.00000*t)-(-0.00000)*sin(15.00000*t))) ).*f_u(t);
```





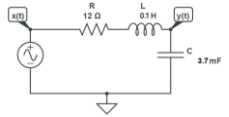
$x(t) = \delta(t)$

$X(s) = 1$

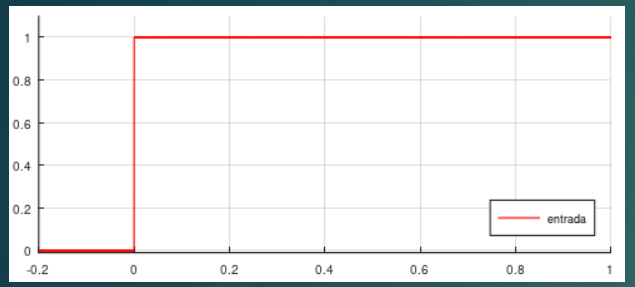
$$H(s) = \frac{2700}{(s+90)(s+30)}$$

$y(0^-) = 0$

$y'(0^-) = 0$



$y(t) = 45e^{-30t} - 45e^{-90t}$



$x(t) = u(t)$

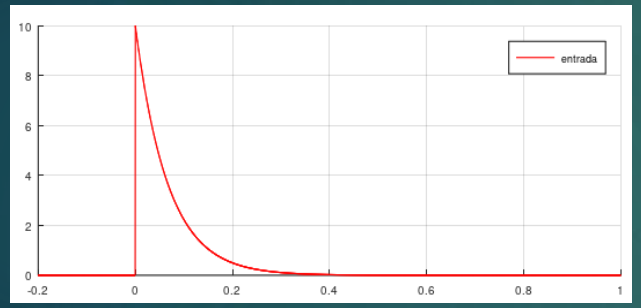
$X(s) = \frac{1}{s}$

$$H(s) = \frac{2700}{(s+90)(s+30)}$$

$y(0^-) = 0$

$y'(0^-) = 0$

$y(t) = 1 - 1.5e^{-30t} + 0.5e^{-90t}$



$x(t) = 10e^{-t}$

$X(s) = \frac{10}{(s+15)}$

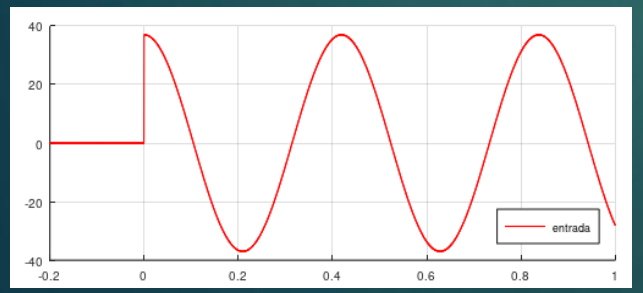
$$H(s) = \frac{2700}{(s+90)(s+30)}$$

$y(0^-) = 5$

$y'(0^-) = 0$

$y_o(t) = 24e^{-15t} - 30e^{-30t} + 6e^{-90t}$

$y(t) = 24e^{-15t} - 22.5e^{-30t} + 3.5e^{-90t}$



$x(t) = 37\cos(15t)$

$X(s) = \frac{37s}{(s^2 + 15^2)}$

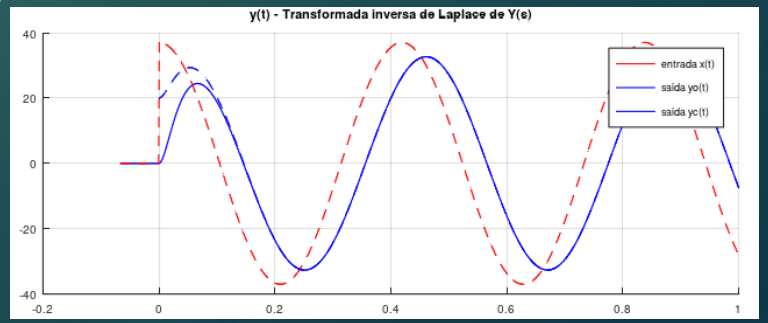
$$H(s) = \frac{2700}{(s+90)(s+30)}$$

$y(0^-) = 20$

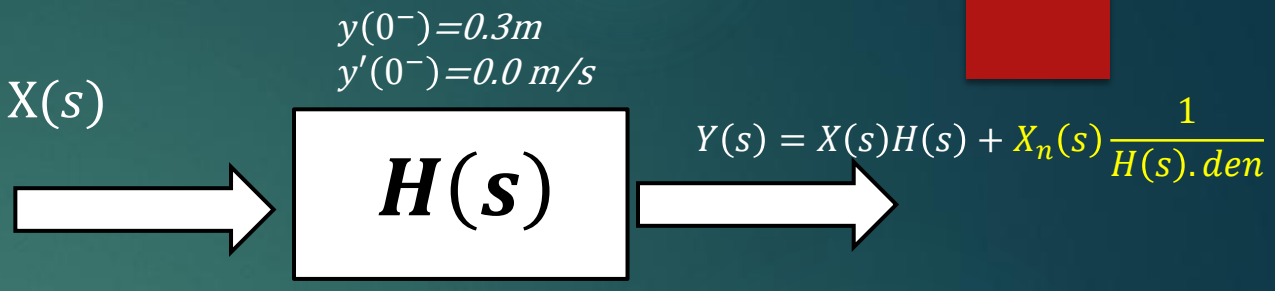
$y'(0^-) = 0$

$y_o(t) = -44.4e^{-30t} + 18e^{-90t} + 26.4\cos(15t) + 19.2\sin(15t)$

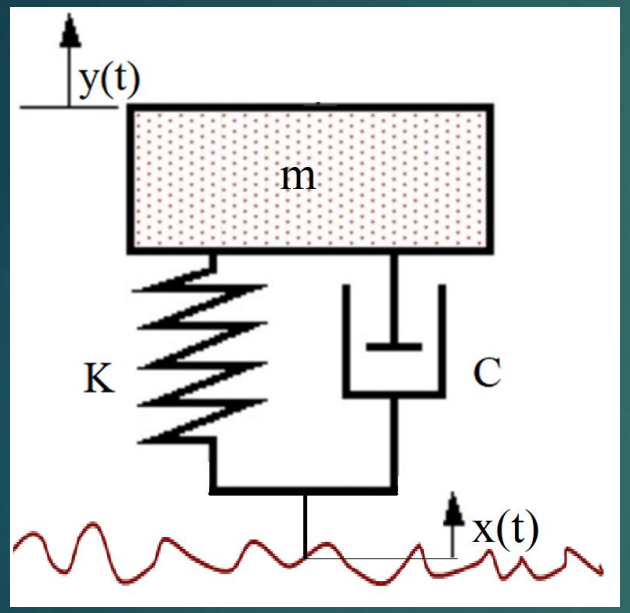
$y(t) = -14.4e^{-30t} + 8.0e^{-90t} + 26.4\cos(15t) + 19.2\sin(15t)$



Sistema mecânico equação diferencial



Sistema



- x(t) será o perfil da estrada
- y(t) será o movimento do carro

$$y''(t) + \frac{C}{m}y'(t) + \frac{K}{m}y(t) = \frac{C}{m}x'(t) + \frac{K}{m}x\ (t)$$

$y(0^-)=0.3m$

$y'(0^-)=0.0\ m/s$

$m=300kg$
$K= 12000\ N/m$
$C=1200\ N/(m/s)$

$$y''(t) + 4y'(t) + 40y(t) = 4x'(t) + 40x(t)$$

Função de Transferência do Sistema

A função de transferência no domínio de Laplace é definida como sendo a razão entre a transformada de Laplace do sinal de saída pela transformada de Laplace do sinal de entrada

Condições Iniciais Nulas

$$y''(t) + 4y'(t) + 40y(t) = 4x'(t) + 40x(t)$$

Aplicacando Laplace em ambos os lados

$$s^2Y(s) + 4sY(s) + 40Y(s) = 4sX(s) + 40X(s)$$

$$(s^2 + 4s + 40)Y(s) = (4s + 40)X(s)$$

$$H(s) = \frac{Y_o(s)}{X(s)} = \frac{4s + 40}{(s^2 + 4s + 40)} = \frac{4s + 40}{(s + 2)^2 + 6^2}$$

$$Y_o(s) = X(s)H(s)$$

$$\boxed{H(s) = \frac{Y(s)}{X(s)} = \frac{4s + 40}{(s^2 + 4s + 40)} = \frac{4s + 40}{(s + 2)^2 + 6^2}}$$

Entrada $x(t) \xrightarrow{\mathcal{L}} X(s)$

$x'(t) \xrightarrow{\mathcal{L}} sX(s)$

Saída $y_o(t) \xrightarrow{\mathcal{L}} Y_o(s)$

$y'(t) \xrightarrow{\mathcal{L}} sY_o(s)$

$$\boldsymbol{H_n(s) = \frac{1}{H(s).den} = \frac{1}{s^2 + as + b}}$$

$$X_n(s) = y(0)(s + a) + y'(0)$$

$y(0^-) = 0$
$y'(0^-) = 0$

$X(s) \Rightarrow \boxed{\boldsymbol{H(s)} = \frac{4s+40}{(s+2)^2+6^2}} \Rightarrow Y_o(s) = X(s)H(s)$

Com Condições Iniciais

$$s^2Y(s) - sy(0) - y'(0) + 4\big(sY(s) - y(0)\big) + 40Y(s) = 4sX(s) + 40X(s)$$

$$(s^2 + 4s + 40)Y(s) - sy(0) - y'(0) - 4y(0) = (4s + 40)X(s)$$

$$Y(s) = X(s)\frac{4s + 40}{(s + 2)^2 + 6^2} + \big(sy(0) + y'(0) + 4y(0)\big)\frac{1}{(s + 2)^2 + 6^2}$$

$$Y(s) = X(s)H(s) + X_n(s)\left(\frac{1}{H(s).den}\right)$$

$$Y(s) = Y_o(s) + X_n(s)\left(\frac{1}{H(s).den}\right)$$

$X(s) \Rightarrow \boxed{\boldsymbol{H(s)} = \frac{4s+40}{(s+2)^2+6^2}} \Rightarrow Y_o(s) = X(s)H(s)$

$y(0^-) = 0.3$
$y'(0^-) = 0$

$X_n(s) = 0.3(s + 4)$

$\Rightarrow \boxed{\boldsymbol{\frac{1}{H(s).den} = \frac{1}{(s + 2)^2 + 6^2}}} \Rightarrow Y_n(s) = X_n(s)\frac{1}{H(s).den}$

Calcular a Resposta ao impulso $h(t)$ do Sistema H(s): $x(t) = \delta(t)$

Condições Iniciais Nulas

$X(s) = 1$ → $\boldsymbol{H(s)} = \frac{4s+40}{(s+2)^2+6^2}$ → $Y(s) = X(s)H(s) = (1)H(s) = H(s)$

Como $x(t) = \delta(t) \xrightarrow{\mathcal{L}} X(s) = 1$ então $Y(s) = X(s)H(s) = (1)H(s) = \frac{4s+40}{(s+2)^2+6^2}$

Então a resposta $h(t)$ do sistema ao impulso será a transformada inversa de H(s)

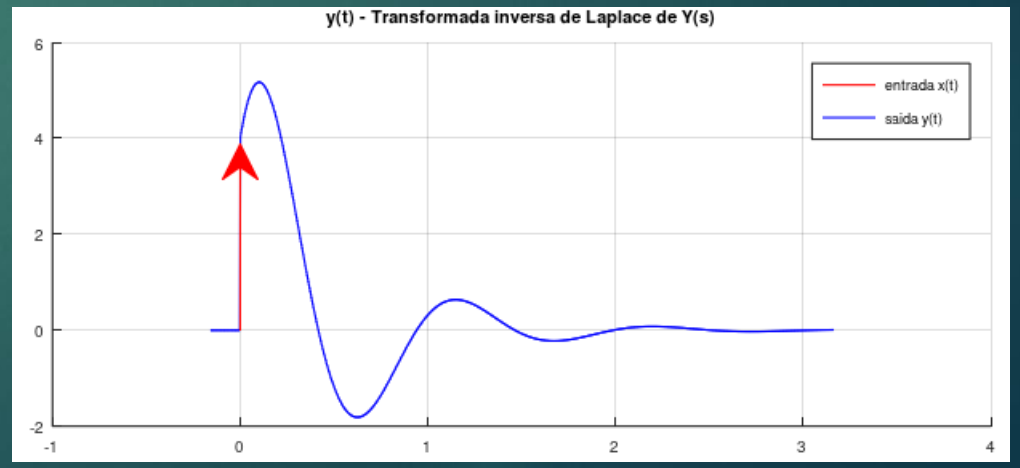
Pólos complexos conjugados $-\sigma \pm jw$ com resíduos $a \pm b = re^{\pm j\theta}$

$$\frac{a+jb}{(s+\sigma-jw)} + \frac{a-jb}{(s+\sigma+jw)} \xRightarrow{\mathcal{L}^{-1}} 2r\,e^{-\sigma t}\cos(wt+\theta) = 2e^{-\sigma t}[a\cos(wt) - b\,sen(wt)]$$

$$H(s) = \frac{4s+40}{(s+2)^2+6^2} = \frac{2-j\frac{8}{3}}{(s+2-j6)} + \frac{2+j\frac{8}{3}}{(s+2+j6)} \xrightarrow{\mathcal{L}^{-1}} h(t) = \left[2e^{-2t}\left(2\cos(6t) + \frac{8}{3}\sin(6t)\right)\right]u(t) = \left[\frac{20}{3}e^{-2t}\cos(6t - 0.9273)\right]u(t)$$

```
>> Hs=(4*s+40)/(s^2+4*s+40);
>> Xs=1;
```

```
>> RespostaSistemaHs(Hs,1,[0 0],2,false);
--Saida--
ft=@(t)(+(2*exp(-2.00000*t).*[2.00000*cos(6.00000*t)-
(-2.66667)*sin(6.00000*t)]) ).*f_u(t);

--Entrada--
fx=@(t)(y=(1.00000)*f_d(t));
```



Calcular a Resposta ao degrau do Sistema H(s): $x(t) = u(t)$

Condições Iniciais Nulas

$$X(s) = \frac{1}{s}$$

$$\boldsymbol{H(s)} = \frac{4s+40}{(s+2)^2+6^2}$$

$$Y(s) = X(s)Y(s) = \left(\frac{1}{s}\right)H(s)$$

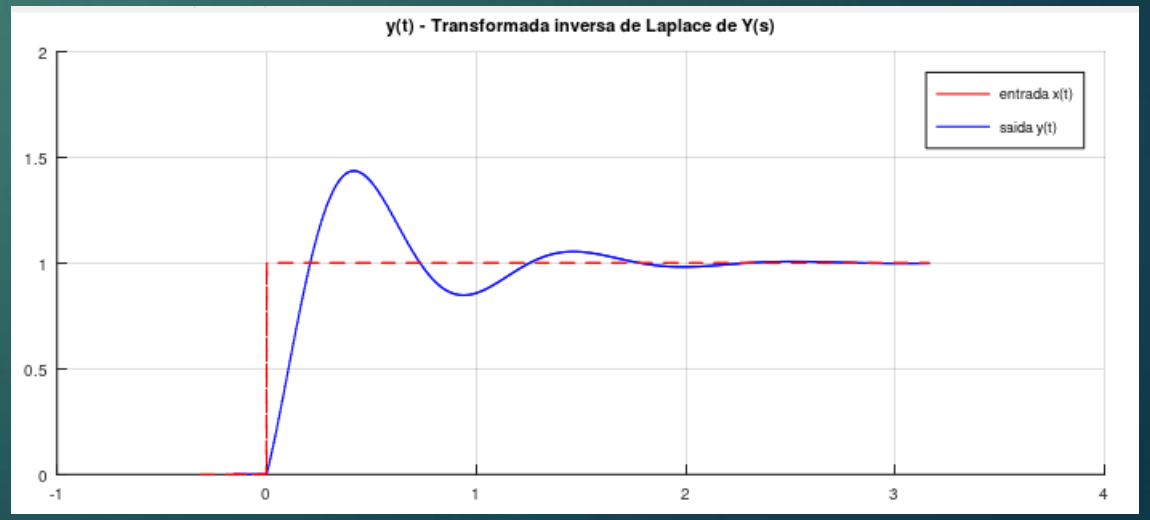
Como $x(t) = u(t) \xrightarrow{\mathcal{L}} X(s) = \frac{1}{s}$ então $Y(s) = X(s)H(s) = \left(\frac{1}{s}\right)H(s) = \frac{1}{s}\frac{4s+40}{\left((s+2)^2+6^2\right)}$

Então a resposta y(t) do sistema será a transformada inversa de Laplace de Y(s), que é o produto da função de transferência pela transformada de Laplace da entrada.

$$Y(s) = X(s)H(s) = \frac{4s+40}{s\left((s+2)^2+6^2\right)} = \frac{1}{s} + \frac{-0.5-j\frac{1}{16}}{(s+2-j6)} + \frac{0.5+j\frac{1}{16}}{(s+2+j6)} \xrightarrow{\mathcal{L}^{-1}} y_u(t) = \left[1 - e^{-2t}\left(\cos(6t) - \frac{1}{3}sen(6t)\right)\right]\mathrm{u(t)}$$

```
>> Hs=(4*s+40)/(s^2+4*s+40);
>> Xs=1/s;
```

```
>> RespostaSistemaHs(Hs,1/s,[0 0],2,true);
--Saida--
ft=@(t)(+(2*(0.52705).*exp(-2.00000*t).*cos(6.00000*t+(-2.81984)))+
(1.00000) ).*f_u(t);

--Entrada--
fx=@(t)(y=(1.00000)*f_u(t)+(0.00000)*f_d(t));
```



Calcular a Resposta do Sistema H(s) à entrada $x(t) = \frac{t}{3}$

$$x(t) = \frac{t}{3} \quad \xrightarrow{X(s) = \frac{(1/3)}{s^2}} \quad \boxed{\boldsymbol{H(s)} = \frac{4s+40}{(s+2)^2+6^2}} \quad \xrightarrow{Y(s) = X(s)Y(s) = \left(\frac{(1/3)}{s^2}\right)H(s)}$$

$y(0^-) = 0.0m$

$y'(0^-) = 0.0\ m/s$

Como $x(t) = \frac{t}{3} \xrightarrow{\mathcal{L}} X(s) = \frac{1/3}{s^2}$ então $Y_0(s) = X(s)H(s) = \frac{1/3}{s^2}\frac{4s+40}{\left((s+2)^2+6^2\right)}$

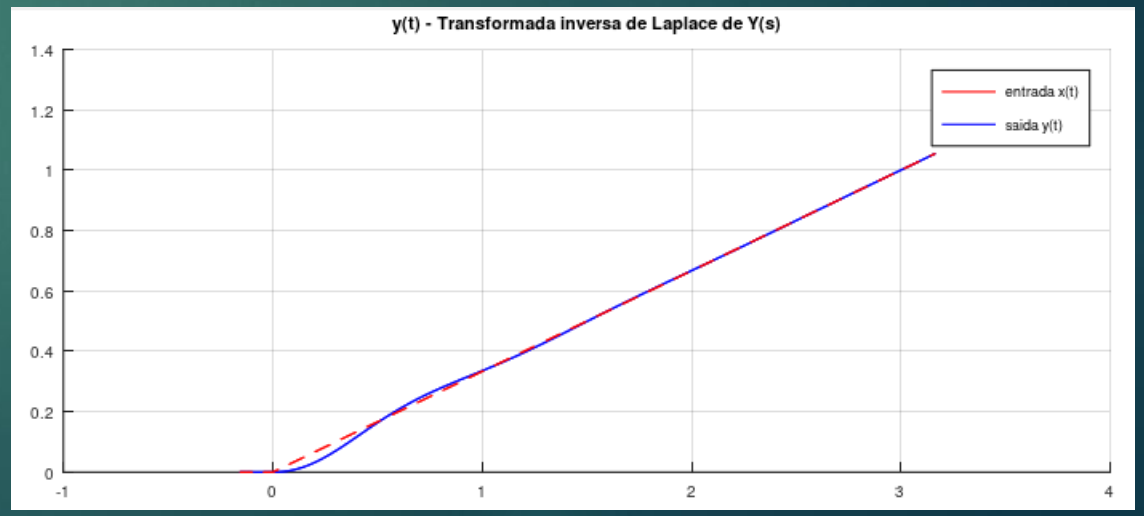
$$Y_0(s) = \frac{\frac{4}{3}s+\frac{40}{3}}{s^2\left((s+2)^2+6^2\right)} = \frac{A}{s} + \frac{B}{s^2} + \frac{Cs+D}{(s+2)^2+6^2} = \frac{0}{s} + \frac{1/3}{s^2} + \frac{-1/3}{(s+2)^2+6^2} = \frac{1/3}{s^2} + \frac{-1/3}{(s+2)^2+6^2}$$

$$Y_0(s) \xRightarrow{\mathcal{L}^{-1}} y_0(t) = \left[\frac{1}{3}t - \frac{1}{18}e^{-2t}sen(6t)\right]u(t)$$

$$\frac{Cs+D}{(s+\sigma)^2+w^2} \xRightarrow{\mathcal{L}^{-1}} e^{-\sigma t}\left(Ccos(wt) + \frac{D-\sigma C}{w}sen(wt)\right)$$

```
>> Hs=(4*s+40)/(s^2+4*s+40);
>> Xs=(1/3)/s^2;
```

```
>> RespostaSistemaHs(Hs,(1/3)/s^2,[0 0],2,true);
--Saida--
ft=@(t)(+(2*(0.02778).*exp(-2.00000*t).*cos(6.00000*t+(1.57080)))+
(0.33333*t) ).*f_u(t);

--Entrada--
fx=@(t)(y=((0.33333)*t+(0.00000)).*f_u(t)+(0.00000)*f_d(t));
```



Calcular a Resposta do Sistema H(s) à entrada $x(t) = 0.377sen(30t)$

$x(t) = 0.377sen(30t)$

$X(s) = \frac{11.31}{(s^2+30^2)}$

$y(0^-)=0.0m$
$y'(0^-)=0.0\ m/s$

$\boldsymbol{H(s)}=\frac{4s+40}{(s+2)^2+6^2}$

$Y_o(s) = X(s)Y(s) = \left(\frac{11.31}{(s^2+30^2)}\right)H(s)$

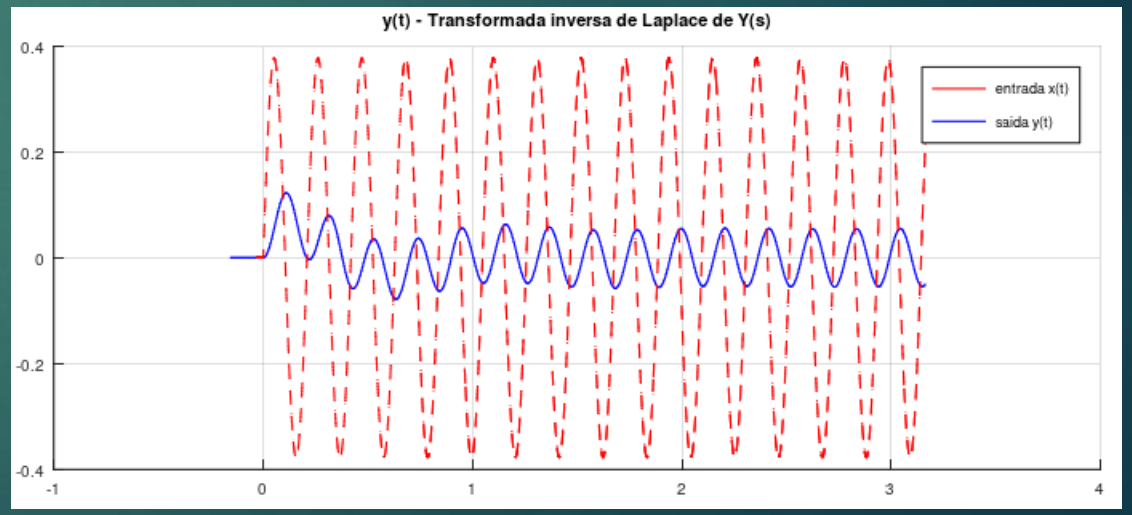
Como $x(t) = 0.377sen(30t) \xrightarrow{\mathcal{L}} X(s) = \frac{11.31}{(s^2+30^2)}$ então $Y_0(s) = X(s)H(s) = \frac{11.31}{(s^2+30^2)}\frac{4s+40}{((s+2)^2+6^2)}$

$$Y_0(s) = \frac{0.027 - j0.034}{(s+2-j6)} + \frac{0.027 + j0.034}{(s+2+j6)} + \frac{-0.027 + j0.005}{(s-j30)} + \frac{-0.027 - j0.005}{(s+j30)}$$

$$Y_0(s) \xRightarrow{\mathcal{L}^{-1}} y_0(t) = (e^{-2t}(0.054\cos(6t) + 0.068sen(6t)) - 0.054\cos(30t) - 0.001sen(30t))u(t)$$

```
>> Hs=(4*s+40)/(s^2+4*s+40);
>> Xs=11.31/(s^2+30^2);
```

```
>> RespostaSistemaHs(Hs,11.31/(s^2+30^2),[0 0],2,true);
--Saida--
ft=@(t)(+(2*(0.02746)*cos(30.00000*t+(2.95848)))+(2*(0.04342).*
exp(-2.00000*t).*cos(6.00000*t+(-0.89965))) ).*f_u(t);

--Entrada--
ft=@(t)(+(2*(0.18850)*cos(30.00000*t+(-1.57080))) ).*f_u(t);
```



Recalcular a Resposta do Sistema H(s) à entrada $x(t) = 0.377sen(30t)$, com condições iniciais

$$x(t) = 0.377sen(30t)$$

$$X(s) = \frac{11.31}{(s^2 + 30^2)}$$

$$y(0^-) = 0.3m$$
$$y'(0^-) = 0.0\ m/s$$

$$\boldsymbol{H(s)} = \frac{4s+40}{(s+2)^2+6^2}$$

$$X_n(s) = 0.3\ (s+4)$$

$$Y(s) = Y_0(s) + Y_n(s) = X(s)H(s) + X_n(s)\frac{1}{H(s).den}$$

$$Y(s) = X(s)H(s) + X_n(s)\frac{1}{H(s).den}$$

$$Y_0(s) = X(s)H(s) = \frac{11.31}{(s^2+30^2)}\ \frac{4s+40}{((s+2)^2+6^2)}$$

$$y_0(t) = (e^{-2t}(0.054\cos(6t) + 0.068sen(6t)) - 0.054\cos(30t) - 0.001sen(30t)\ )u(t)$$

Como $y(0) = 0.3\ e\ \ y'(0) = 0\ \ \rightarrow X_n(s) = 0.3s + 1.2$

$$Y(s) = Y_o(s) + (0.3s + 1.2)\left(\frac{1}{(s+2)^2 + 6^2}\right)$$

$$0.3\ (s+4)\frac{1}{(s+2)^2+6^2} \xrightarrow{\mathcal{L}^{-1}} y_n(t) = e^{-2t}(0.3\cos(6t) + 0.1sen(6t))$$

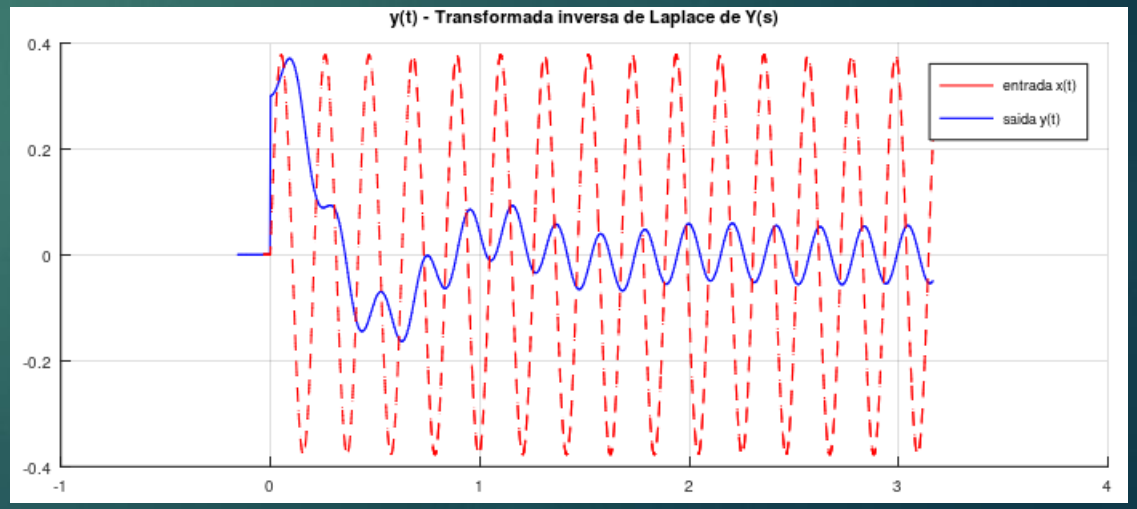
$$y\ (t) = y_o(t) + y_n(t) = (e^{-2t}(0.354\cos(6t) + 0.168sen(6t)) - 0.054\cos(30t) - 0.001sen(30t)\ )u(t)$$

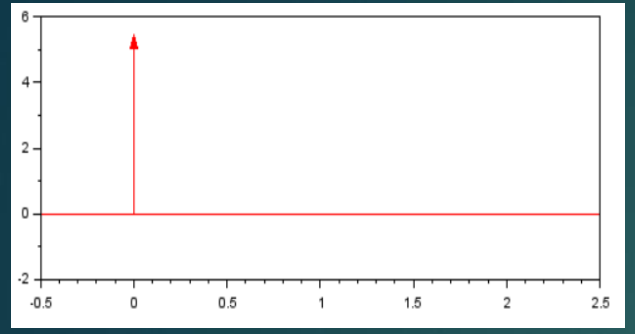
$$\boldsymbol{H_n(s)} = \frac{\mathbf{1}}{\boldsymbol{H(s).den}} = \frac{\mathbf{1}}{\boldsymbol{s^2 + as + b}}$$
$$X_n(s) = y(0)(s+a) + y'(0)$$

$$a = 4;\quad y(0) = 0.3;\quad y'(0) = 0$$
$$X_n(s) = 0.3\ (s+4)$$

```
>> Hs=(4*s+40)/(s^2+4*s+40);
>> Xs=11.31/(s^2+30^2);
>> [num_c,den_c] = tfdata(Hs,'v');
>> Hn=tf(1,den_c);
>> Xn=EntradaNula(Hs,[0.3,0]);
>> ft=ilaplace_fga(Hs*Xs+Hn*Xn,true)
ft=@(t)(+(2*(0.02746)*cos(30.00000*t+(2.95848)))+(2*(0.19592).*exp(-2.00000*t).*
cos(6.00000*t+(-0.44310))) ).*f_u(t);
```

```
>> RespostaSistemaHs(Hs,11.31/(s^2+30^2),[0.3 0],2,true);
--Saida--
ft=@(t)(+(2*(0.02746)*cos(30.00000*t+(2.95848)))+(2*(0.19592).*exp(-2.00000*t).*
cos(6.00000*t+(-0.44310))) ).*f_u(t);

--Entrada--
ft=@(t)(+(2*(0.18850)*cos(30.00000*t+(-1.57080))) ).*f_u(t);
```





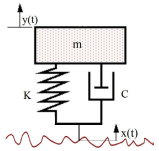
$x(t) = \delta(t)$

$X(s) = 1$

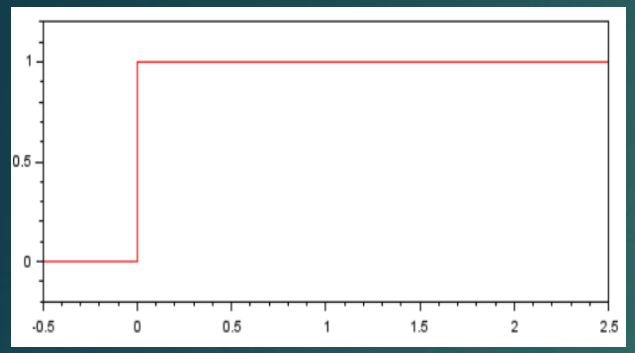
$$\boldsymbol{H(s)} = \frac{4s + 40}{(s+2)^2 + 6^2}$$

$y(0^-) = 0.0$

$y'(0^-) = 0.0$



$$h(t) = e^{-2t}\left(4\cos(6t) + \frac{16}{3}sin(6t)\right)$$



$x(t) = u(t)$

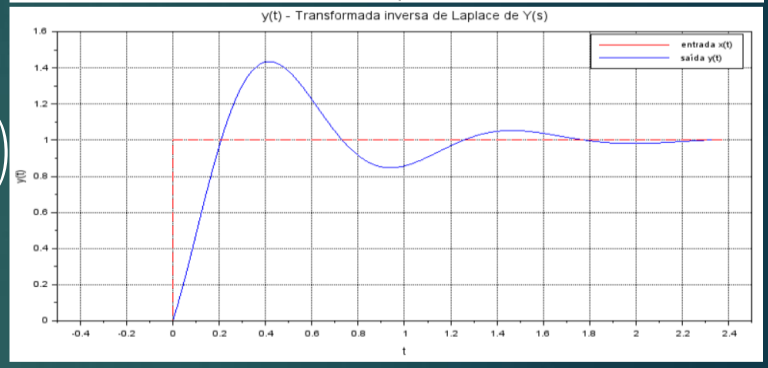
$X(s) = \frac{1}{s}$

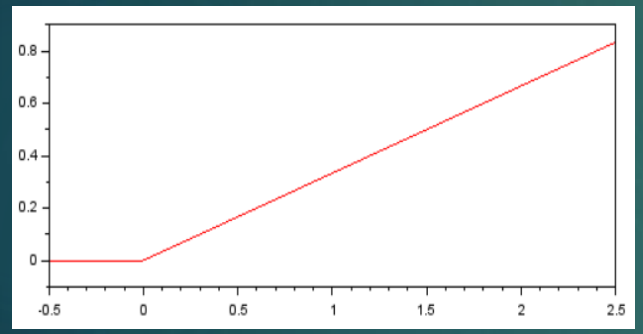
$$\boldsymbol{H(s)} = \frac{4s + 40}{(s+2)^2 + 6^2}$$

$y(0^-) = 0.0$

$y'(0^-) = 0.0$

$$y_u(t) = 1 - e^{-2t}\left(\cos(6t) - \frac{1}{3}sen(6t)\right)$$





$x(t) = \frac{t}{3}$

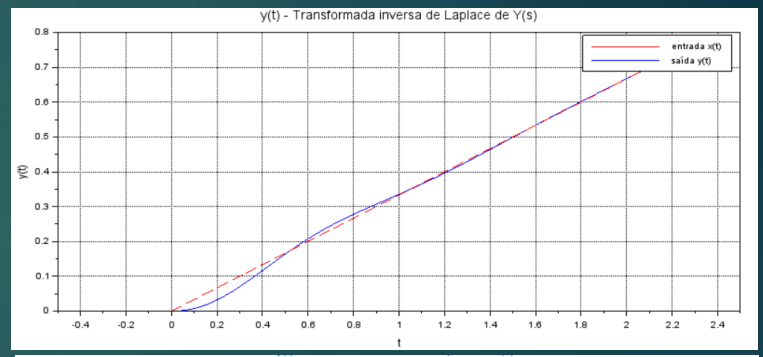
$X(s) = \frac{1/3}{s^2}$

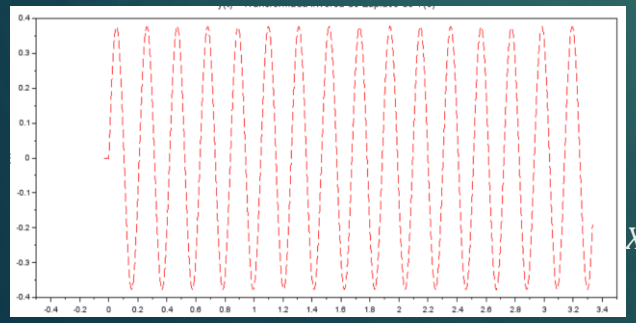
$$\boldsymbol{H(s)} = \frac{4s + 40}{(s+2)^2 + 6^2}$$

$y(0^-) = 0.0$

$y'(0^-) = 0.0$

$$y_o(t) = \frac{1}{3}\,t - \frac{1}{18}e^{-2t}sen(6t)$$





$x(t) = 0.1sen(30t)$

$X(s) = \frac{11.31}{(s^2 + 30^2)}$

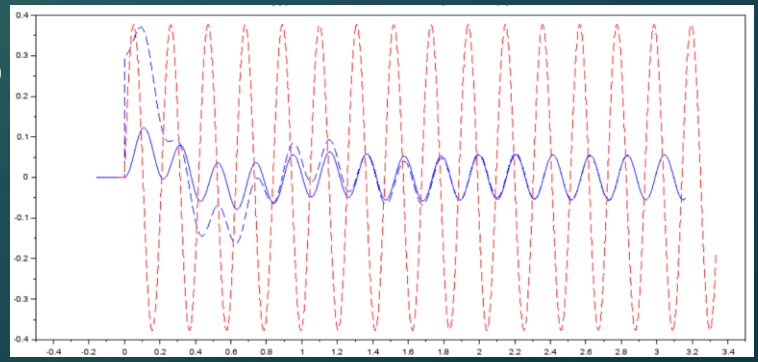
$$\boldsymbol{H(s)} = \frac{4s + 40}{(s+2)^2 + 6^2}$$

$y(0^-) = 0.3$

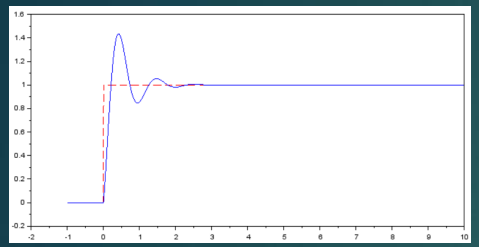
$y'(0^-) = 0.0$

$$y_0(t) = e^{-2t}(0.054\cos(6t) + 0.068\,sen(6t)) - 0.054\cos(30t) - 0.001\text{sen}(30t$$

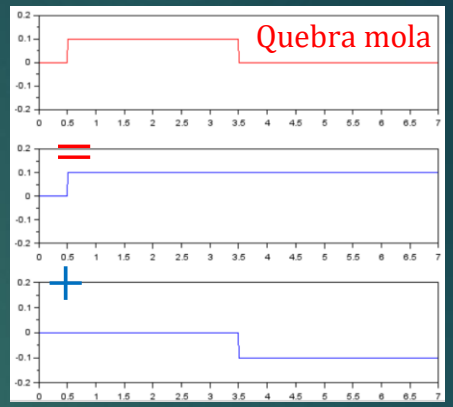
$$y(t) = e^{-2t}(0.354\cos(6t) + 0.168\text{sen}(6t)) - 0.054\cos(30t) - 0.001\text{sen}(30t)$$



Calcular a resposta deste Sistema um Quebra Mola $de\ largura\ T = 3, centrado\ em\ t = 2, com\ altura\ 0.1$



$$y_u(t) = \left[1 - e^{-2t}\left(\cos(6t) - \frac{1}{3}sen(6t)\right)\right]\mathrm{u(t)}$$



$$x(t) = 0.1\left(u(t-0.5) - u(t-3.5)\right)$$

Como já conhecemos a resposta ao degrau $y_u(t) = 1 - e^{-2t}\left(\cos(6t) - \frac{1}{3}sen(6t)\right)$, podemos usar a linearidade e a invariância no tempo:

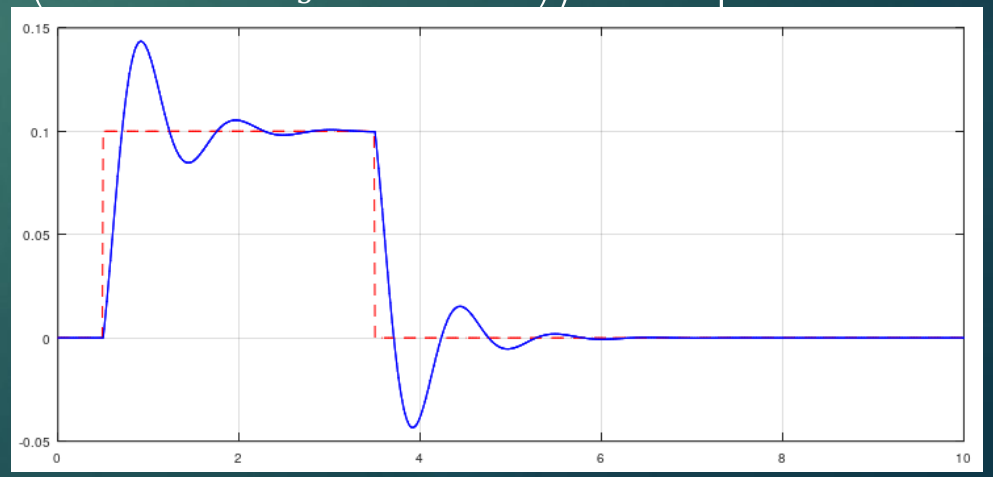
$$x(t) = 0.1\left(u(t-0.5) - u(t-3.5)\right) \Rightarrow \boxed{\boldsymbol{H(s)} = \frac{4s+40}{(s+2)^2+6^2}} \Rightarrow y(t) = 0.1\left(y_u(t-0.5) - y_u(t-3.5)\right)$$

$$X(s) = 0.1\frac{e^{-0.5s} - e^{-3.5s}}{s} \qquad Y(s) = 0.1\left(\frac{e^{-0.5s} - e^{-3.5s}}{s}\right)\frac{6s+45}{(s^2+6s+45)}$$

$$y(t) = 0.1\left[\left(1 - e^{-2(t-0.5)}\left(\cos\left(6(t-0.5)\right) - \frac{1}{3}sen\left(6(t-0.5)\right)\right)\right)\mathrm{u(t-0.5)} - \left(1 - e^{-2(t-3.5)}\left(\cos\left(6(t-3.5)\right) - \frac{1}{3}sen\left(6(t-3.5)\right)\right)\right)\mathrm{u(t-3.5)}\right]$$

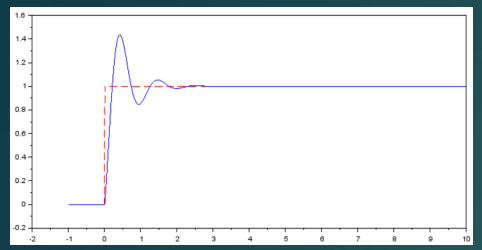
```
>> Hs=(4*s+40)/(s^2+4*s+40);
>> Xs=1/s;
>> ft=ilaplace_fga(Hs*Xs,true);
ft=@(t)(+(2*(0.52705).*exp(-2.00000*t).*cos(6.00000*t+(-2.81984)))+
(1.00000) ).*f_u(t);
```

```
>> t=linspace(0,10,1001);
>> hold on; grid on
>> plot(t,0.1*(f_u(t-0.5)-f_u(t-3.5)),'r--');
>> plot(t,0.1*(ft(t-0.5)-ft(t-3.5)),'b');
```

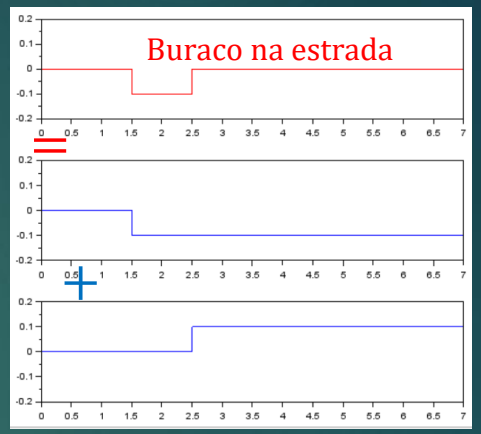


Calcular a resposta deste Sistema um Buraco na Estrada

$$de\ largura\ T = 1, centrado\ em\ t = 2, com\ altura\ \ k = -0.1$$



$$y_u(t) = \left[1 - e^{-2t}\left(\cos(6t) - \frac{1}{3}sen(6t)\right)\right]\mathrm{u(t)}$$



$$x(t) = -0.1\left(u\left(t - 2 + \frac{T}{2}\right) - u\left(t - 2 - \frac{T}{2}\right)\right)$$

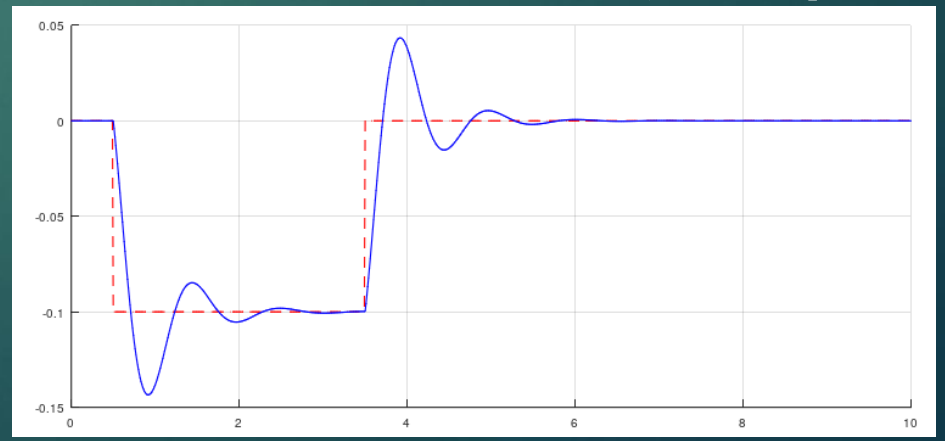
Como já conhecemos a resposta ao degrau $y_u(t) = 1 - e^{-2t}\left(\cos(6t) - \frac{1}{3}sen(6t)\right)$, podemos usar a linearidade e a invariância no tempo:

$$x(t) = -0.1\big(u(t-1.5) - u(t-2.5)\big) \Rightarrow \boxed{\boldsymbol{H(s)} = \frac{4s+40}{(s+2)^2+6^2}} \Rightarrow y(t) = -0.1\big(y_u(t-1.5) - y_u(t-2.5)\big)$$

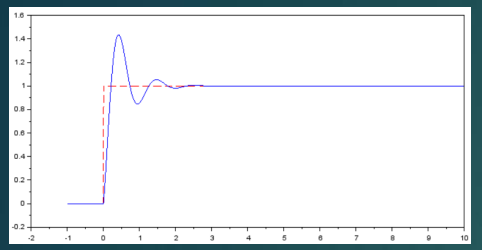
$$X(s) = -0.1\frac{e^{-1.5s} - e^{-2.5s}}{s} \qquad Y(s) = 0.1\left(\frac{e^{-1.5s} - e^{-2.5s}}{s}\right)\frac{6s+45}{(s^2+6s+45)}$$

$$y(t) = -0.1\left[\left(1 - e^{-2(t-1.5)}\left(\cos\big(6(t-1.5)\big) - \frac{1}{3}sen\big(6(t-1.5)\big)\right)\right)\mathrm{u(t-1.5)} - \left(1 - e^{-2(t-2.5)}\left(\cos\big(6(t-2.5)\big) - \frac{1}{3}sen\big(6(t-2.5)\big)\right)\right)\mathrm{u(t-2.5)}\right]$$

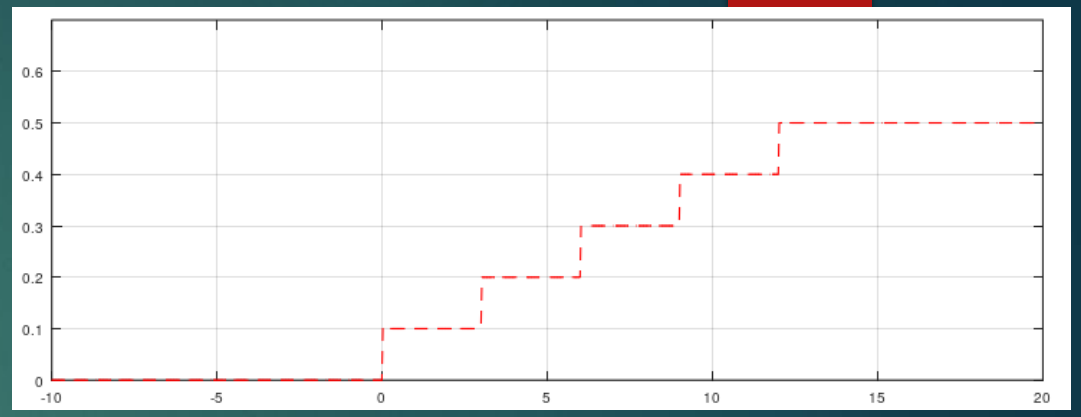
```
>> t=linspace(0,10,1001);
>> hold on; grid on
>> plot(t,-0.1*(f_u(t-0.5)-f_u(t-3.5)),'r--');
>> plot(t,-0.1*(ft(t-0.5)-ft(t-3.5)),'b');
```



Calcular a resposta deste Sistema : subida de uma escada de 5 degraus $de\ largura\ T = 3, com\ altura\ k = 0.1$



$$y_u(t) = \left[1 - e^{-2t}\left(\cos(6t) - \frac{1}{3}sen(6t)\right)\right]\mathrm{u(t)}$$



$$x(t) = 0.1\ (u(t) + u(t-3) + u(t-6) + u(t-9) + u(t-12))$$

```
>> t=linspace(-10,20,1001);
>> plot(t,0.1*(f_u(t)+f_u(t-3)+f_u(t-6)+f_u(t-9)+f_u(t-12)),'r--');
```

Como já conhecemos a resposta ao degrau $y_u(t) = 1 - e^{-2t}\left(\cos(6t) - \frac{1}{3}sen(6t)\right)$, podemos usar a linearidade e a invariância no tempo:

$$x(t) = 0.1\begin{pmatrix} u(t) + u(t-3) \\ +u(t-6) + u(t-9 \end{pmatrix} + u(t-12))$$

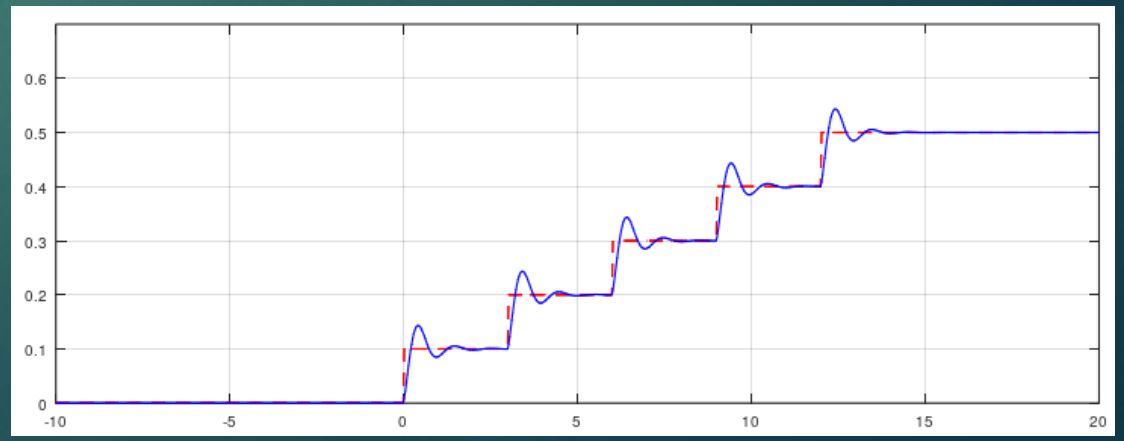
$$\boldsymbol{H(s)} = \frac{4s + 40}{(s+2)^2 + 6^2}$$

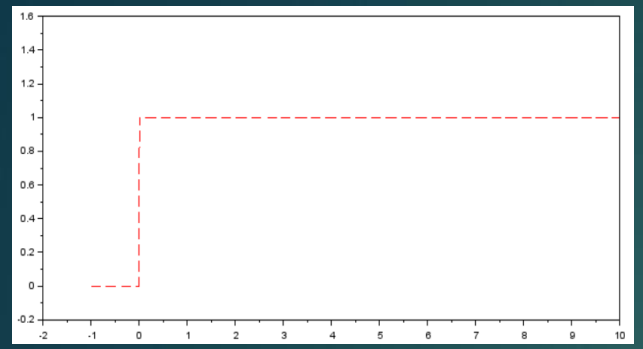
$$y(t) = 0.1\ (y_u(t) + y_u(t-3) + y_u(t-6) + y_u(t-9) + y_u(t-12))$$

$$X(s) = -0.1\frac{1 + e^{-3s} - e^{-6s} + e^{-9s} - e^{-12s}}{s}$$

$$Y(s) = -0.1\left(\frac{1 + e^{-3s} - e^{-6s} + e^{-9s} - e^{-12s}}{s}\right)\frac{6s + 45}{(s^2 + 6s + 45)}$$

```
>> t=linspace(-10,20,1001);
>> grid on; hold on;
>> plot(t,0.1*(f_u(t)+f_u(t-3)+f_u(t-6)+f_u(t-9)+f_u(t-12)),'r--');
>> plot(t,0.1*(ft(t)+ft(t-3)+ft(t-6)+ft(t-9)+ft(t-12)),'b');
```





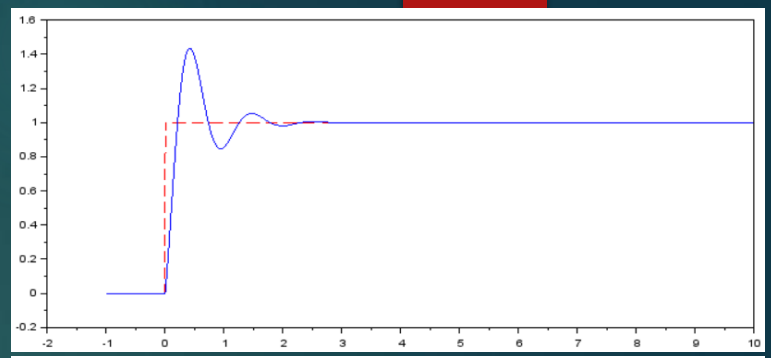
$$x(t) = u(t)$$

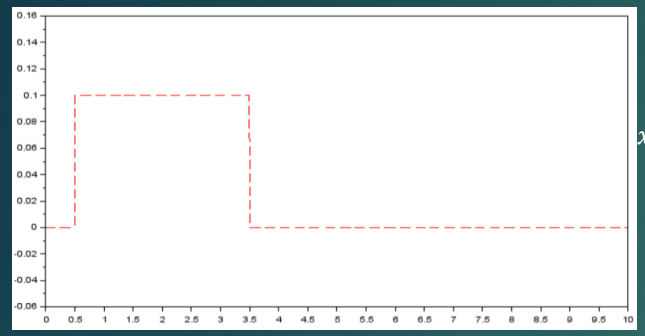
$$X(s) = \frac{1}{s}$$

$$\boldsymbol{H(s)} = \frac{4s + 40}{(s+2)^2 + 6^2}$$

$y(0^-)=0.0$
$y'(0^-)=0.0$

$$y_u(t) = 1 - e^{-2t}\left(\cos(6t) - \frac{1}{3}sen(6t)\right)$$





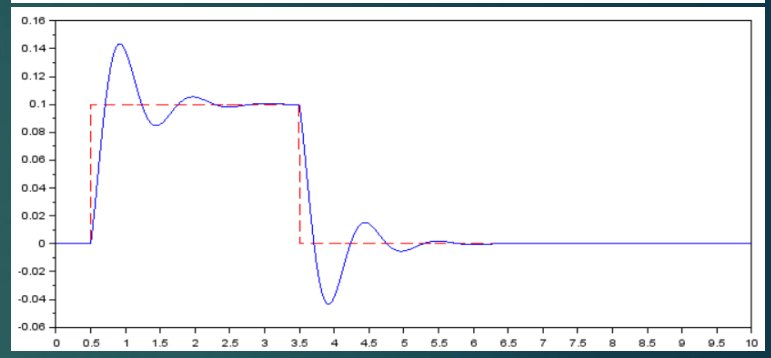
$$x(t) = 0.1\big(u(t-0.5) - u(t-3.5)\big)$$

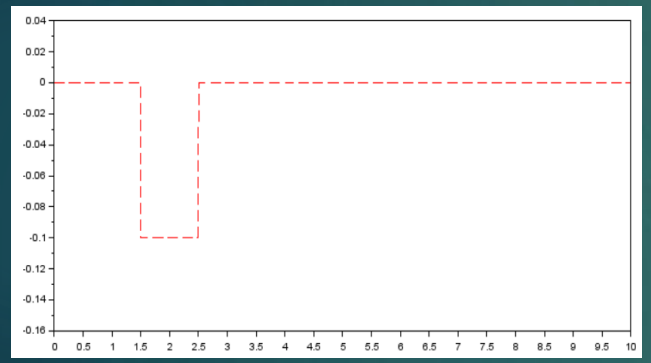
$$X(s) = 0.1\left(\frac{e^{-0.5s} - e^{-3.5s}}{s}\right)$$

$$\boldsymbol{H(s)} = \frac{4s + 40}{(s+2)^2 + 6^2}$$

$y(0^-)=0.0$
$y'(0^-)=0.0$

$$y(t) = 0.1\big(y_u(t-0.5) - y_u(t-3.5)\big)$$





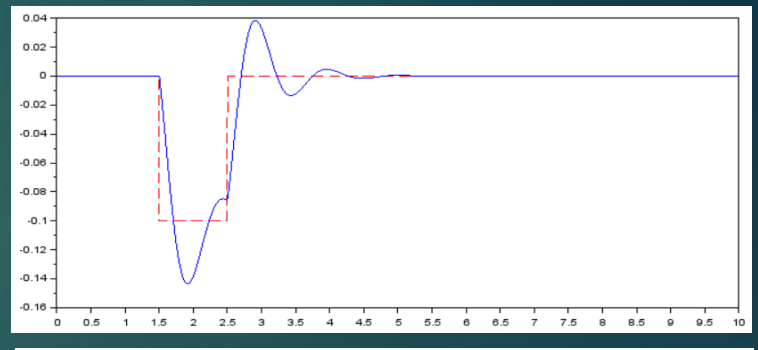
$$x(t) = -0.1\big(u(t-1.5) - u(t-2.5)\big)$$

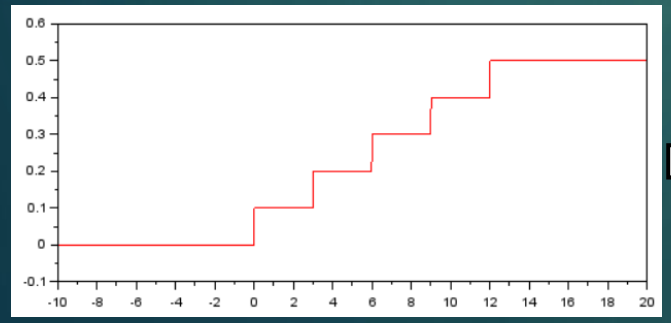
$$X(s) = -0.1\left(\frac{e^{-1.5s} - e^{-2.5s}}{s}\right)$$

$$\boldsymbol{H(s)} = \frac{4s + 40}{(s+2)^2 + 6^2}$$

$y(0^-)=0.0$
$y'(0^-)=0.0$

$$y(t) = -0.1\big(y_u(t-1.5) - y_u(t-2.5)\big)$$





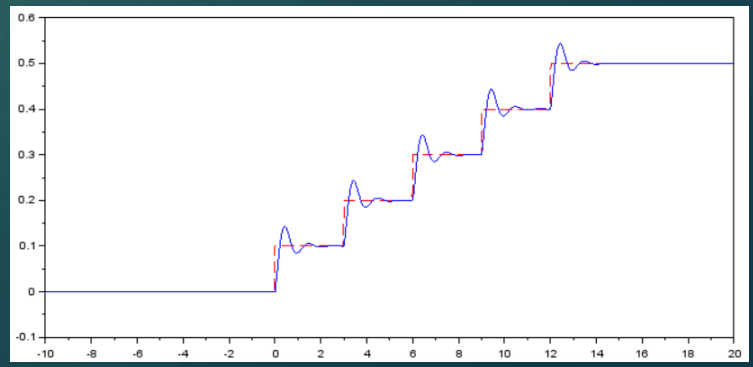
$$x(t) = 0.1\,(u(t) + u(t-3) + u(t-6) + u\,(t-9) + u(t-12))$$

$$X(s) = -0.1\left(\frac{1 + e^{-3s} - e^{-6s} + e^{-9s} - e^{-12s}}{s}\right)$$

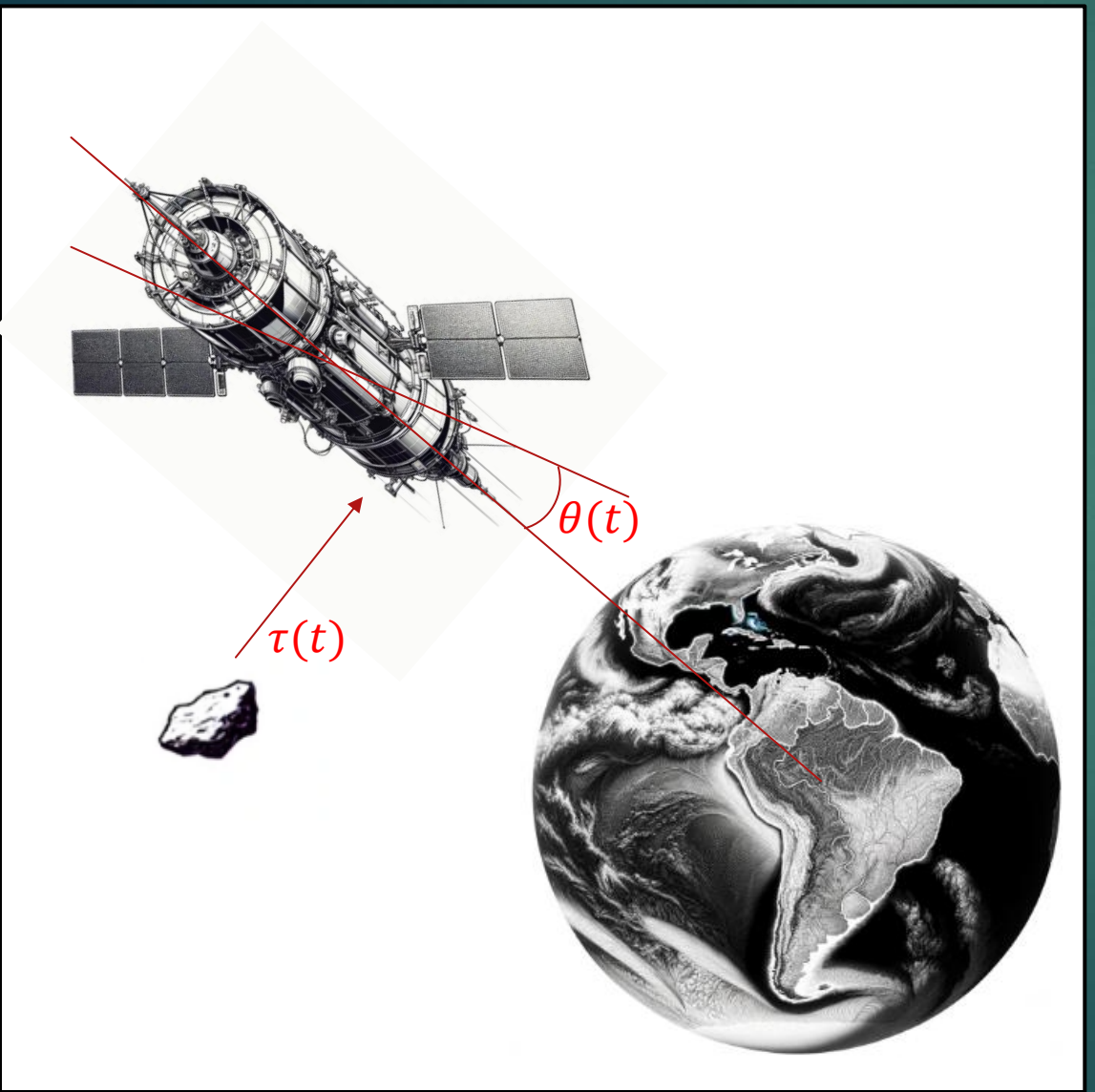
$$\boldsymbol{H(s)} = \frac{4s + 40}{(s+2)^2 + 6^2}$$

$y(0^-)=0.0$
$y'(0^-)=0.0$

$$y(t) = 0.1\,(y_u(t) + y_u(t-3) + y_u(t-6) + y_u(t-9) + y_u(t-12))$$



Sistema Aeroespacial - Equação diferencial



$y(0^-)=0.0m$
$y'(0^-)=0.0\ m/s$

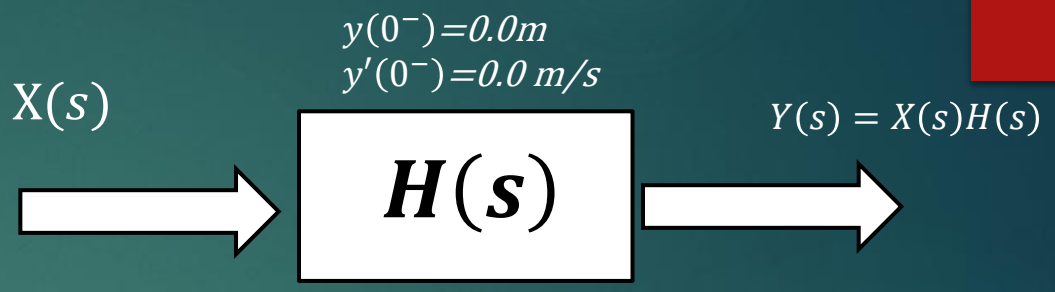
$X(s)$ → $\boldsymbol{H(s)}$ → $Y(s) = X(s)H(s)$

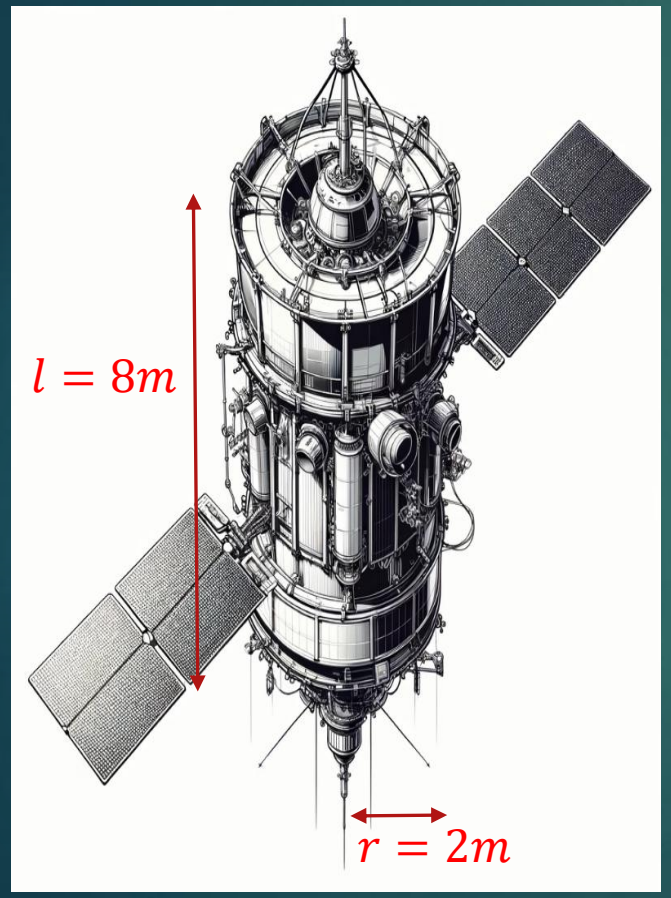
- Entrada: $\tau(t)$ será um torque que tire o satélite do eixo gravitacional
- Saída: $\theta(t)$ será o consequente movimento de rotação do satélite em relação ao eixo gravitacional

$$I_d\ \theta''(t) + b\theta'(t) + K_g\theta(t) = \tau(t)$$

- I_d - Momento de inércia do Satélite
- K_g - Rigidez gravitacional
- b = coeficiente de amortecimento

Sistema Aeroespacial - Momento de Inércia (I)





O momento de inércia é uma medida da tendência de um corpo a resistir à aceleração angular em torno de um eixo de rotação.

$$I = \int r^2 dm$$

No caso de um cilindro com dimensões $l_s = 8m$, $r_s = 1m$, $e\ massa\ \mathrm{m_s} = 400\mathrm{kg}$ (e considerando que a massa esteja uniformemente distribuída no volume do satélite), teremos a seguinte simplificação:

$$I_l = \frac{1}{2} m_s r_s^2 = \frac{1}{2} 400\ 1^2 = 200\ kg.m^2$$

$$I_r = \frac{1}{12} m_s (3r_s^2 + l_s^2) = \frac{1}{12} 400(3\ 1^2 + 8^2) = 2233\ kg.m^2$$

Sistema Aeroespacial - Coeficiente de Rigidez Gravitacional K_g

K_g: Coeficiente de rigidez gravitacional, que relaciona o torque restaurador ao desvio angular θ, ajudando a trazer o satélite de volta à sua orientação de equilíbrio.

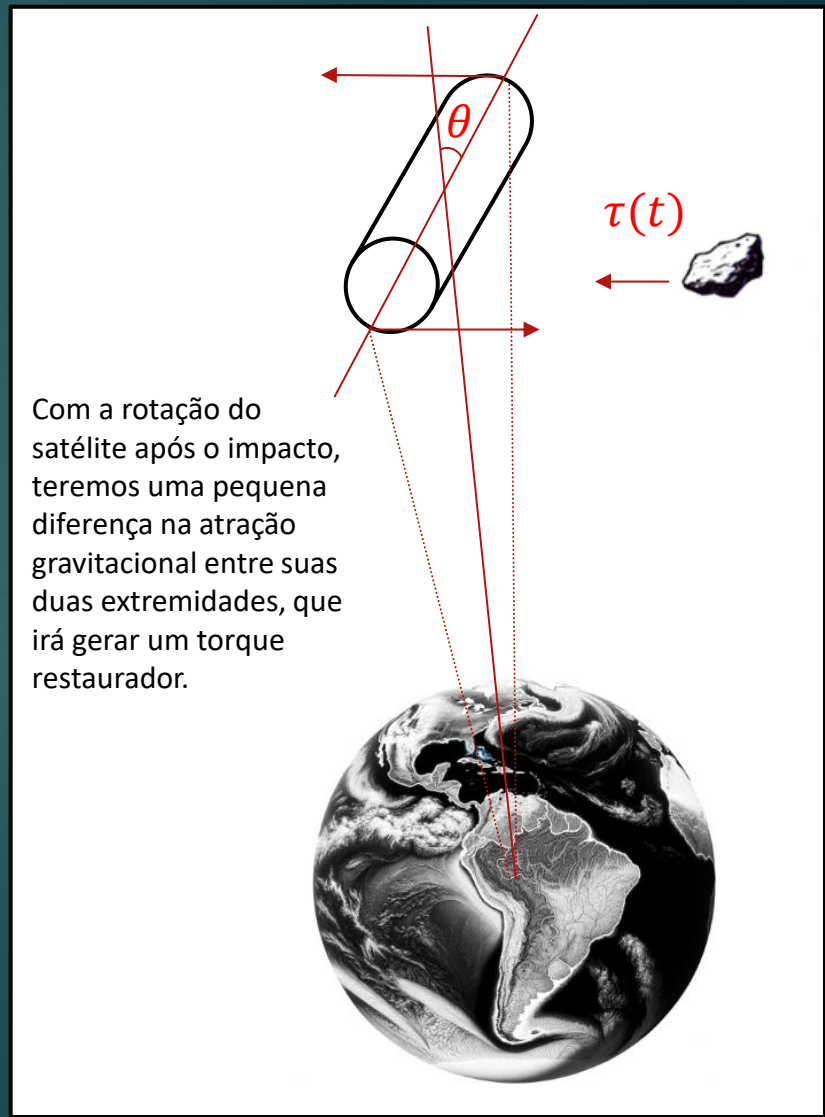
$$K_g = 3\,\frac{G\,m_t}{r_o^3}\Delta I$$

$$G\ - Constante\ Gravitacional\ Universal = 6.674\ 10^{-11}\,\frac{m^3}{kg.s^2}$$

$$r_t\ - Raio\ da\ terra\ = 6371km$$
$$m_t\ - Massa\ da\ terra = 5.972\ 10^{24}\ kg$$
$$h\ - altitude\ do\ satélite = 400km$$
$$r_o -\ \ raio\ da\ órbita\ = r_T + h = 6371 + 400 = 6771km$$

$$\Delta I = I_r - I_l = 2233 - 200 = 2033\ kg.m^2$$

$$K_g = 3\,\frac{G\,m_t}{r_o^3}\Delta I = 3 * 6.674\ 10^{-11} * 5.972\ 10^{24} * \frac{2033}{6771^3} = 0.00783\ \text{N·m/rad.}$$



Com a rotação do satélite após o impacto, teremos uma pequena diferença na atração gravitacional entre suas duas extremidades, que irá gerar um torque restaurador.

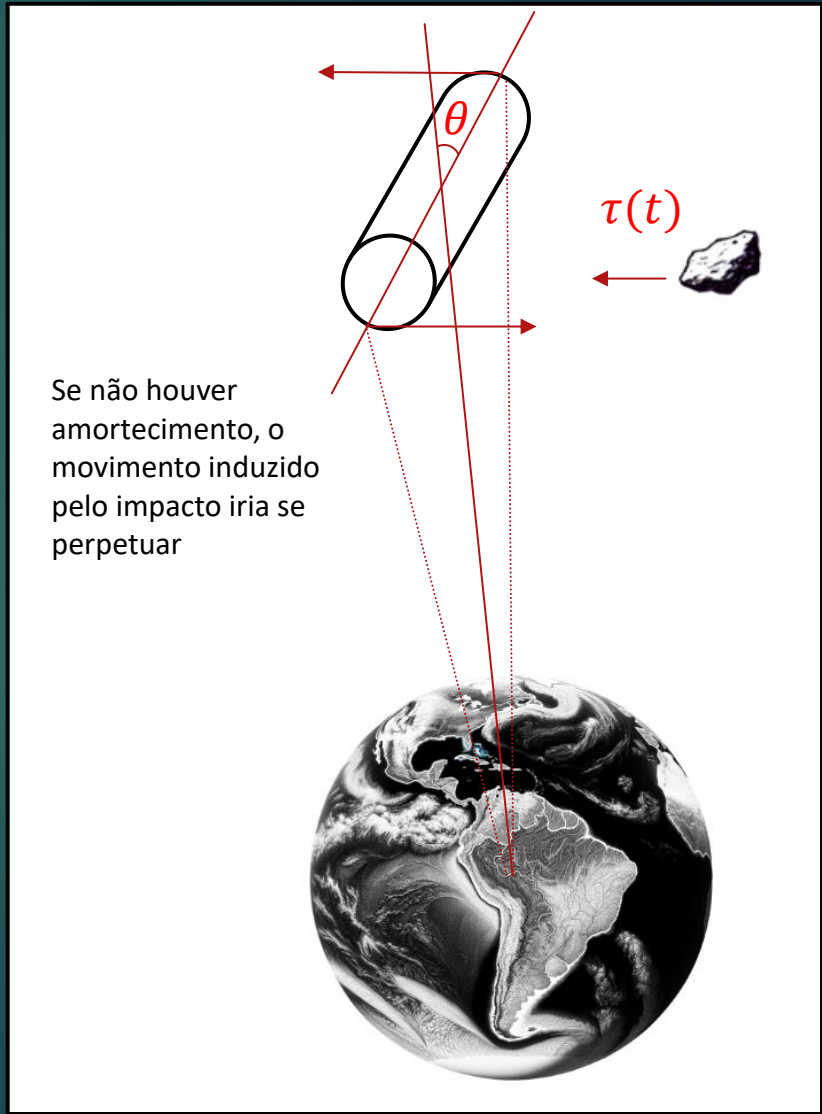
Sistema Aeroespacial - Coeficiente de Amortecimento b

b: Coeficiente de amortecimento, que atua sobre a velocidade angular θ'(t), representando a resistência ao movimento rotacional. Este termo é crucial para modelar como o movimento rotacional do satélite é reduzido ao longo do tempo devido ao amortecimento.

Há várias opções para o amortecimento, o mais simples é a Roda de Reação Passiva. Ela consiste simplesmente de uma roda ou volante em um eixo de rotação livre. Quando o satélite sofre uma perturbação (como uma colisão com lixo espacial), a roda de reação começa a girar devido à conservação do momento angular, criando um momento angular oposto ao da perturbação. Ela gradualmente desacelera devido ao atrito interno de suas engrenagens. Essa desaceleração dissipa o momento angular indesejado gerado pela perturbação.

O valor de b depende do atrito nas engrenagens da roda de reação.

b=1 N.m.s/rad.



Sistema Aeroespacial - Coeficiente de Amortecimento b

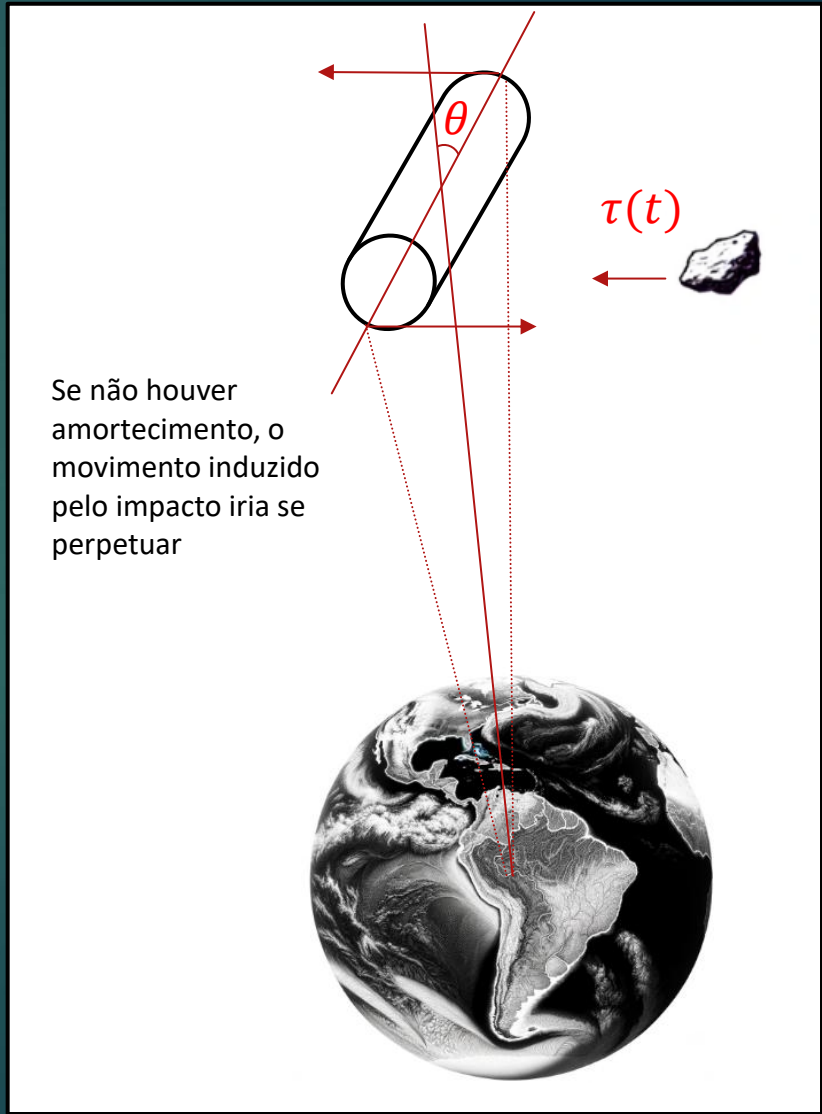
b: Coeficiente de amortecimento, que atua sobre a velocidade angular θ'(t), representando a resistência ao movimento rotacional. Este termo é crucial para modelar como o movimento rotacional do satélite é reduzido ao longo do tempo devido ao amortecimento.

Outra opção seria o sistema o magnético passivo. Para isso , o satélite precisa ser equipado com barras ou fitas de materiais que têm uma alta permeabilidade magnética e uma baixa coercividade, de modo a serem capazes de se magnetizar e desmagnetizar facilmente em resposta a um campo magnético externo, que no caso seria o campo magnético da terra.

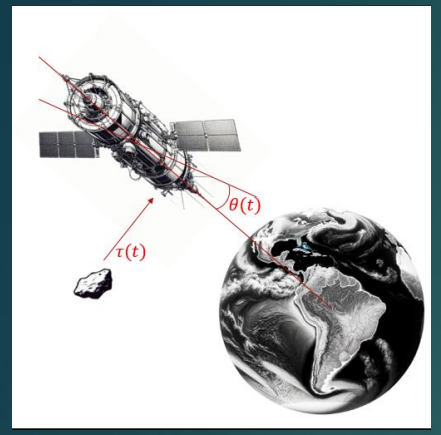
Quando o satélite sofre uma perturbação (como uma colisão com lixo espacial), mudando assim sua orientação em relação ao campo magnético da Terra, correntes de Foucault são induzidas nas barras magnéticas dentro do satélite que, por sua vez, geram campos magnéticos opostos aos que as induziram, dissipando assim energia.

O valor de b depende dos ímãs utilizado, mas seria da ordem de:

b=1 N.m.s/rad.



Sistema Aeroespacial - Resolução por Laplace



$$I_d\,\theta''(t) + b\theta'(t) + K_g\theta(t) = \tau(t)$$

$$\theta''(t) + \frac{b}{I_d}\theta'(t) + \frac{K_g}{I_d}\theta(t) = \frac{1}{I_d}\tau(t)$$

$$\boldsymbol{I} = \mathbf{200}\ N.m.s^2/rad$$
$$\boldsymbol{b} = \mathbf{1.0}\ N.m.s/rad$$
$$\boldsymbol{K_g} = \mathbf{0.00783}\ \ N.m/rad$$
$$\theta''(t) + 0.005\,\theta' + 3.915\ 10^{-5}\,\theta(t) = \tau(t)$$

$\theta(0^-) = 0.0m$
$\theta'(0^-) = 0.0\ m/s$

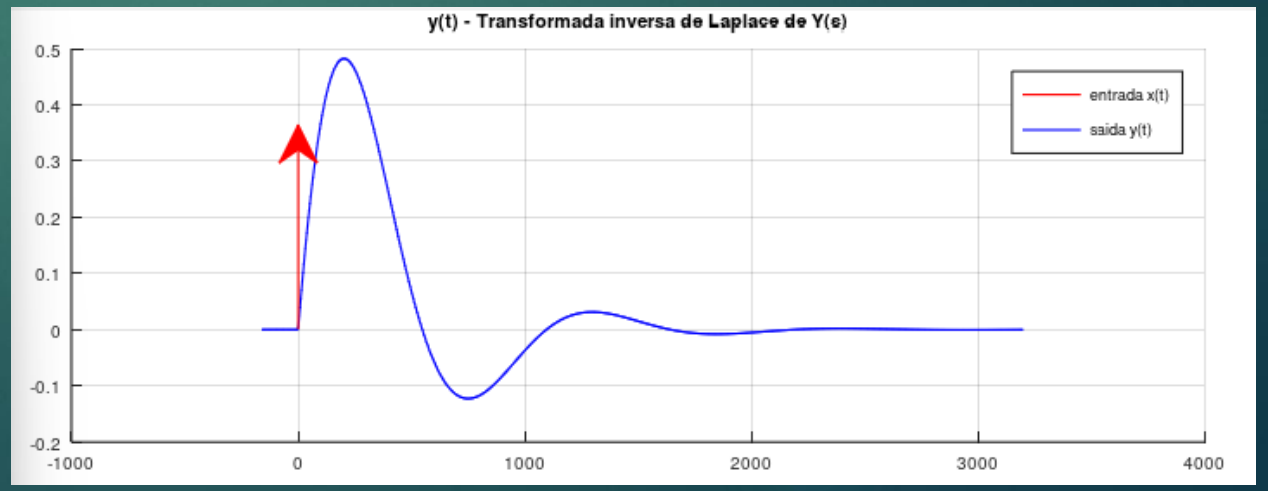
$$\mathrm{T}(s) \longrightarrow \boxed{\boldsymbol{H(s)}} \longrightarrow \Theta\,(s) = \mathrm{T}(s)H(s)$$

$$H(s) = \frac{\Theta\,(s)}{\mathrm{T}(s)} = \frac{0.005}{s^2 + 0.005s + 3.915\ 10^{-5}} = \frac{0.005}{(s+0.0025)^2 + 0.00574^2}$$

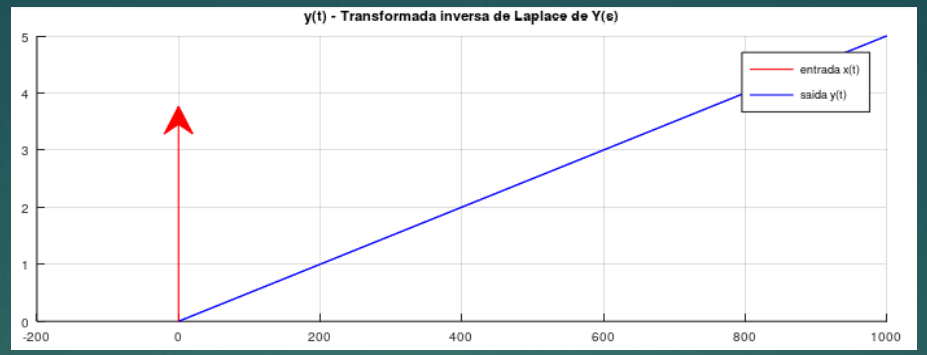
```
>> Hs = 0.005/((s+0.0025)^2 +0.00574^2);
>> RespostaSistemaHs(Hs,1,0,2,1)
--Saida--
ft=@(t)(+(2*(0.43554).*exp(-0.00250*t).*cos(0.00574*t+(-1.57080))) ).*f_u(t);

--Entrada--
fx=@(t)(y=(1.00000)*f_d(t));
```

O impacto da pedra pode ser modelado por um impulso ,
Então $\tau(t) = \delta(t)$

Notar que após a perturbação, o sistema necessitará de aproximadamente 33 min (2000 s) para se estabilizar. E o ângulo máximo de desvio θ será de 28 graus (0.49 radianos)



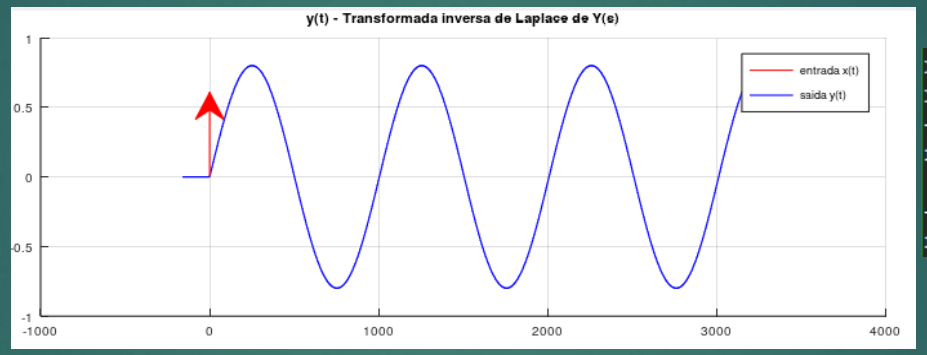
$$\boldsymbol{I} = \mathbf{200}\ N.m.s^2/rad$$
$$\boldsymbol{b} = \mathbf{0}$$
$$\boldsymbol{K_g} = \mathbf{0}$$
$$\theta''(t) + \frac{0}{200}\theta'(t) + \frac{0}{200}\theta(t) = \frac{1}{200}\tau(t)$$
$$H(s) = \frac{0.005}{s^2}$$




```
>> Hs=0.005/s^2;
>> RespostaSistemaHs(Hs,1,0,2,1)
--Saida--
ft=@(t)(+(0.00500*t) ).*f_u(t);

--Entrada--
fx=@(t)(y=(1.00000)*f_d(t));
```

Deep-space – Hs caso não haja nem gravidade ($\boldsymbol{K_g} = \mathbf{0}$) nem campo magnético (ou roda de reação) ($\boldsymbol{b} = \mathbf{0}$), após o choque o satélite iria girar indefinidamente.

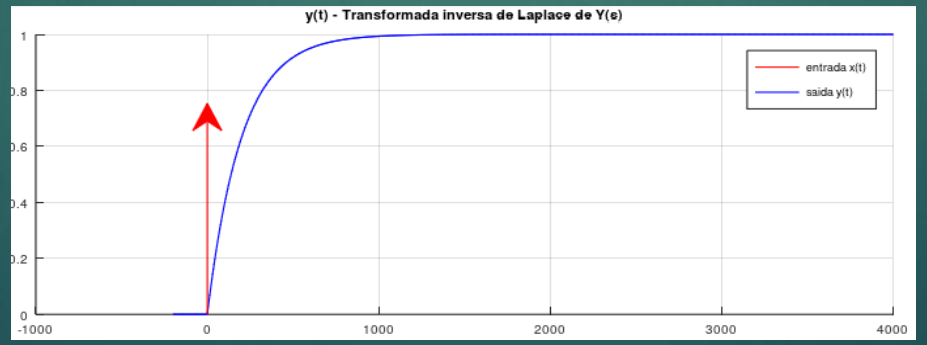
$$\boldsymbol{I} = \mathbf{200}\ N.m.s^2/rad$$
$$\boldsymbol{b} = \mathbf{0}\ \ N.m.s/rad$$
$$\boldsymbol{K_g} = \mathbf{0.00783}\ \ N.m/rad$$
$$\theta''(t) + \frac{0}{200}\theta'(t) + \frac{\mathbf{0.00783}}{200}\theta(t) = \frac{1}{200}\tau(t)$$
$$H(s) = \frac{0.005}{s^2 + 3.915\ 10^{-5}} = \frac{0.005}{s^2 + 0.006257^2}$$




```
>> Hs=0.005/(s^2+0.006257^2);
>> RespostaSistemaHs(Hs,1,0,2,1)
--Saida--
ft=@(t)(+(2*(0.39955)*cos(0.00626*t+(-1.57080))) ).*f_u(t);

--Entrada--
fx=@(t)(y=(1.00000)*f_d(t));
```

Caso não campo magnético ou roda de reação($\boldsymbol{b} = \mathbf{0}$), após o choque o satélite iria oscilar, girando 45° (0.8 rad) em uma direção e depois 45° em outra indefinidamente.

$$\boldsymbol{I} = \mathbf{200}\ N.m.s^2/rad$$
$$\boldsymbol{b} = \mathbf{1.0}\ N.m.s/rad$$
$$\boldsymbol{K_g} = \mathbf{0}\ N.m/rad$$
$$\theta''(t) + \frac{1}{200}\theta'(t) + \frac{0}{200}\theta(t) = \frac{1}{200}\tau(t)$$
$$H(s) = \frac{0.005}{s^2 + 0.005s}$$




```
>> Hs=0.005/(s^2 +0.005*s);
>> RespostaSistemaHs(Hs,1,0,2,1)
--Saida--
ft=@(t)(+(-1.00000*exp(-0.00500*t))+(1.00000) ).*f_u(t);

--Entrada--
fx=@(t)(y=(1.00000)*f_d(t));
```

Deep-space - Caso não haja gravidade ($\boldsymbol{K_g} = \mathbf{0}$), após o choque o satélite iria girar e se estabilizar em uma nova posição rotacionado de 57° (1.0 rad).